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(54) **VECTORS FOR DELIVERY OF HUMAN GROWTH HORMONE GENE AND MULTIPLE THERAPEUTIC GENES INTO CENTRAL NERVOUS SYSTEM BY CROSSING BLOOD BRAIN BARRIER**

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CPC *A61K 48/0058* (2013.01); *A61K 9/0019* (2013.01); *A61K 38/27* (2013.01); *A61K 48/0075* (2013.01); *C12N 15/86* (2013.01); *C12N 15/88* (2013.01); *C12N 2750/14143* (2013.01); *C12N 2830/008* (2013.01)

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Publication Classification

(51) **Int. Cl.**

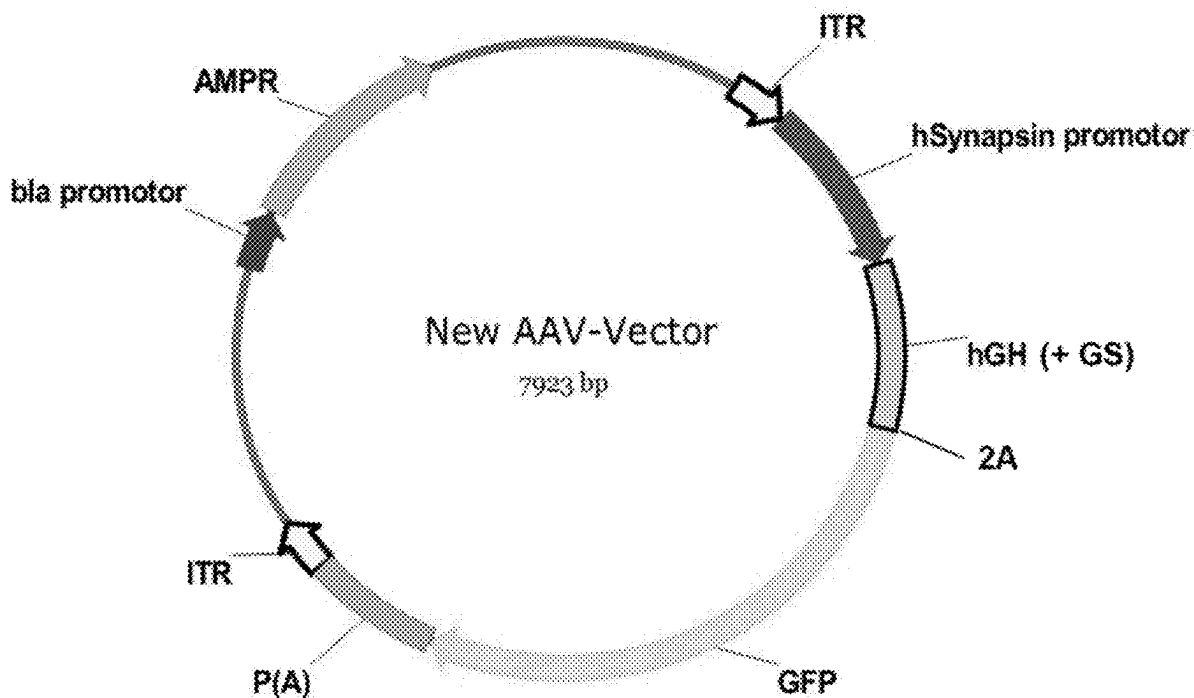
A61K 48/00 (2006.01)

A61K 9/00 (2006.01)

(57) **ABSTRACT**

The present disclosure provides an expression vector and a method for generating the expression vector. The expression vector includes one or more inserted foreign DNA sequences, including human growth hormone (hGH) and a plurality of therapeutic genes, where the one or more inserted foreign DNA sequences are fused with intracellular or extracellular secretion sequences; a promoter, configured to drive expression of the one or more inserted foreign DNA sequences; an insulator, configured for stable and safety expression of the one or more inserted foreign DNA sequences; a marker gene, configured to encoding one or more corresponding proteins to indicate the expression of the one or more inserted foreign DNA sequences; and a regulator element, configured to control expression of the hGH and the plurality of therapeutic genes under a controllable condition.

Specification includes a Sequence Listing.



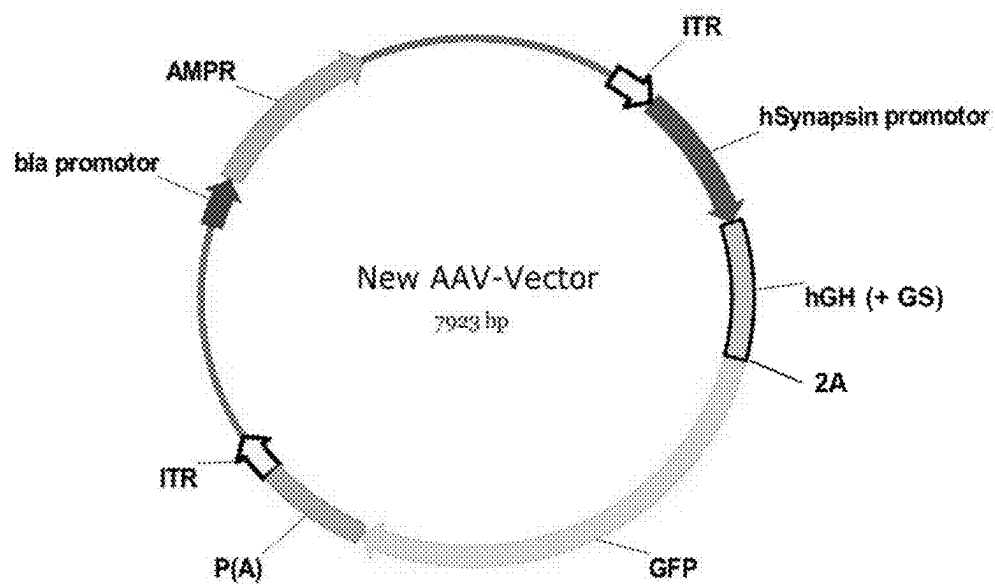


FIG. 1

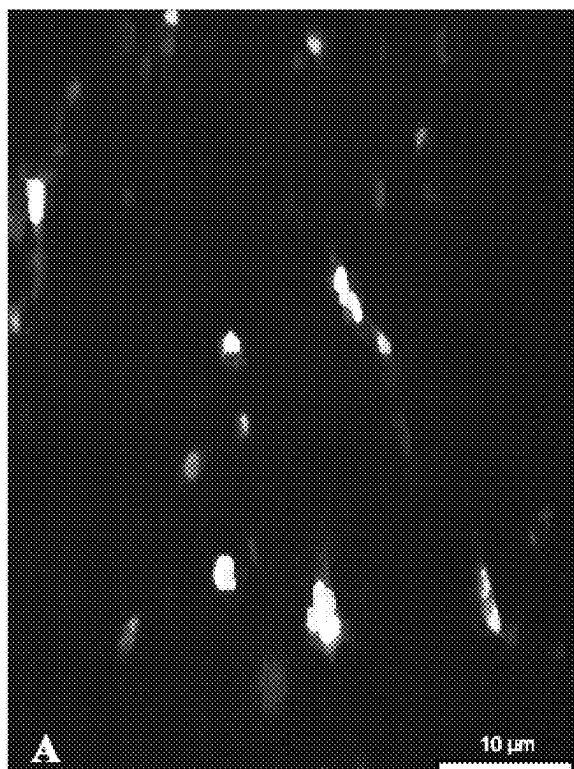


FIG. 2A

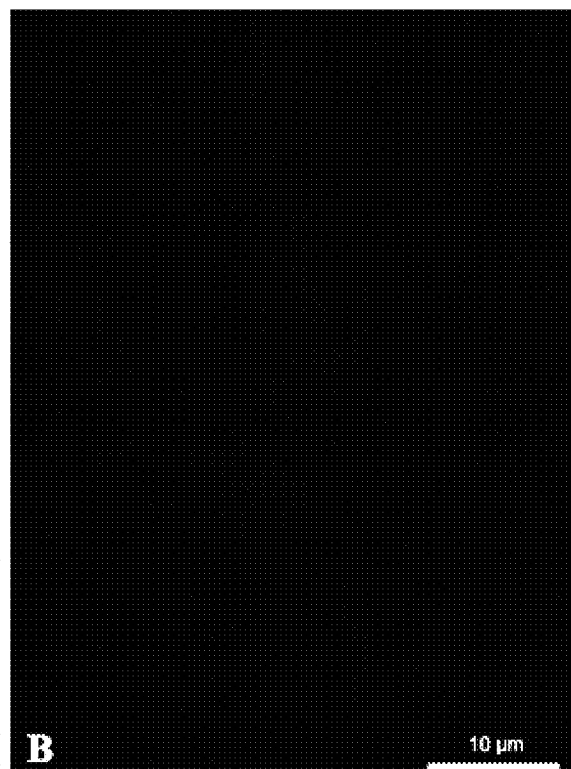


FIG. 2B

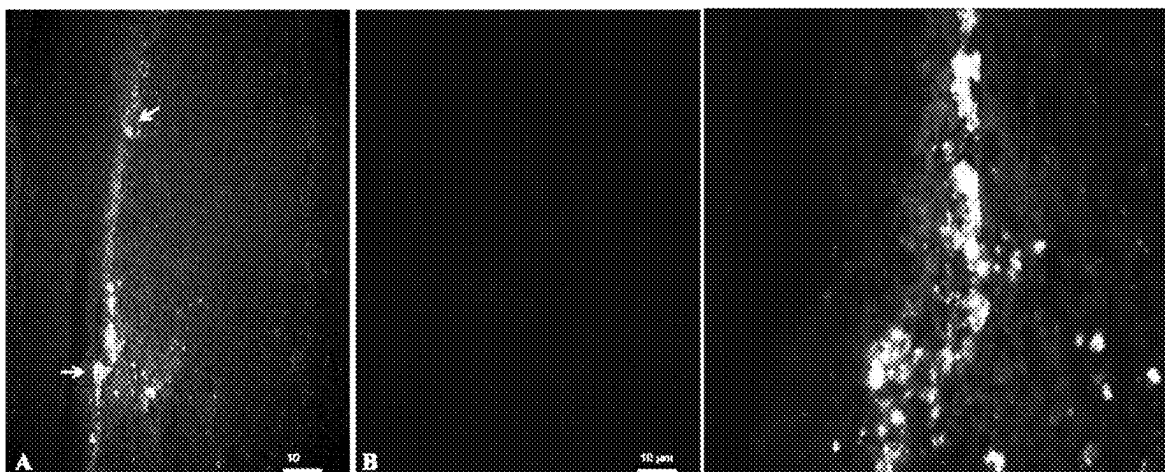


FIG. 3A

FIG. 3B

FIG. 3C

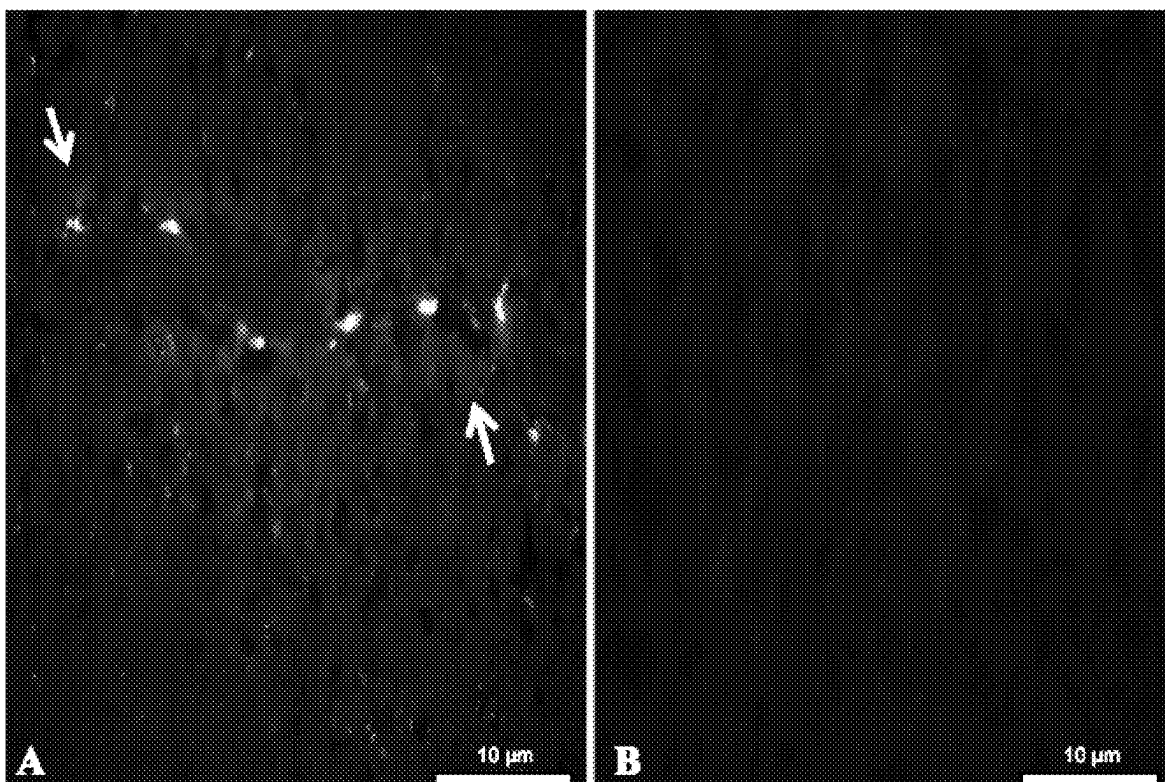


FIG. 4A

FIG. 4B

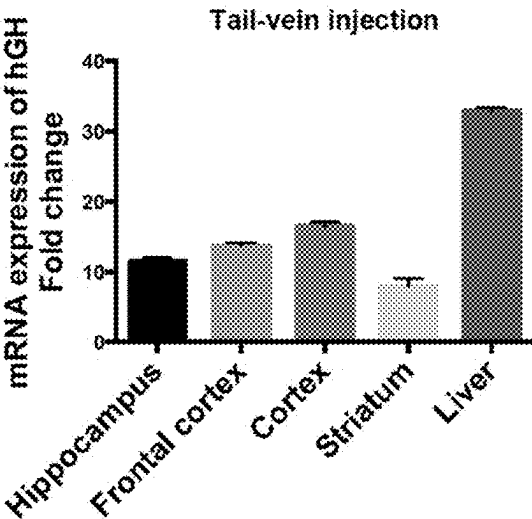


FIG. 5A

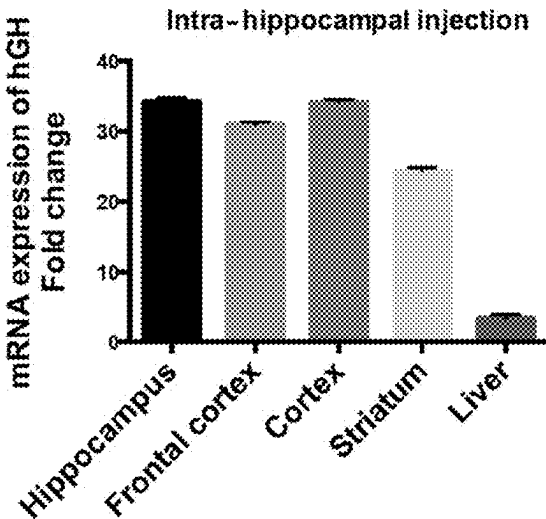


FIG. 5A

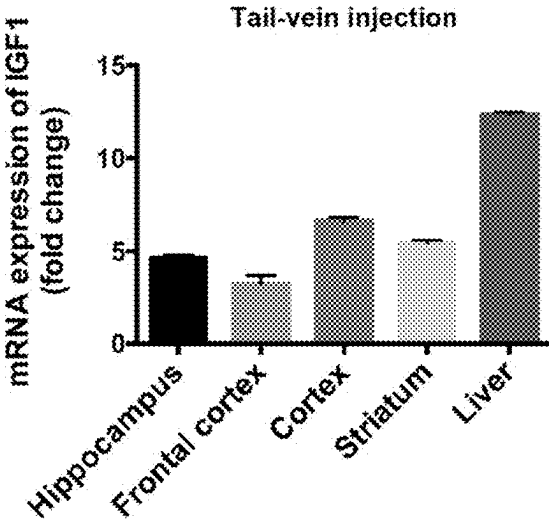


FIG. 6A

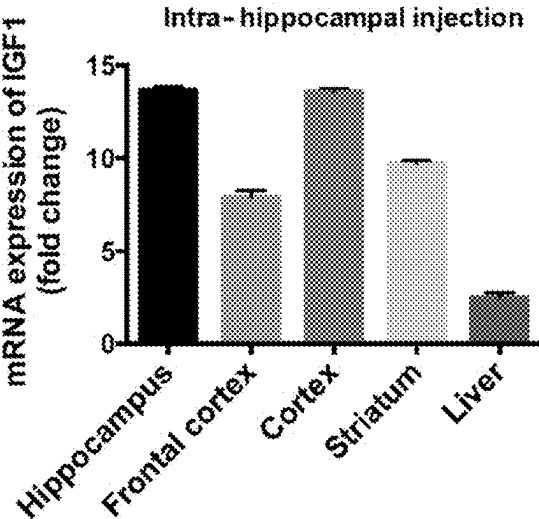


FIG. 6B

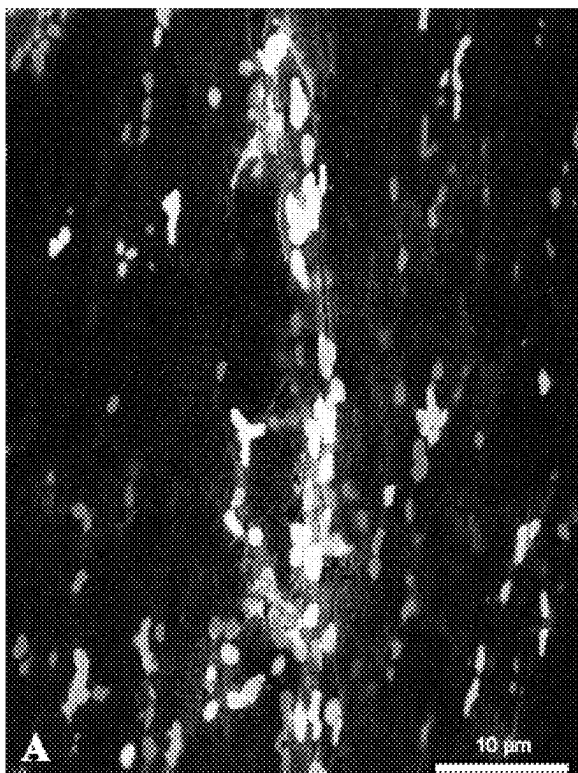


FIG. 7A

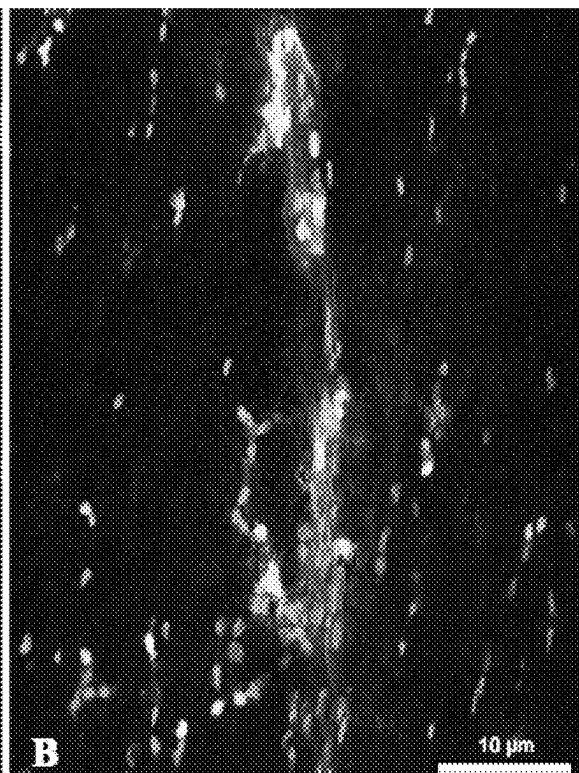


FIG. 7B

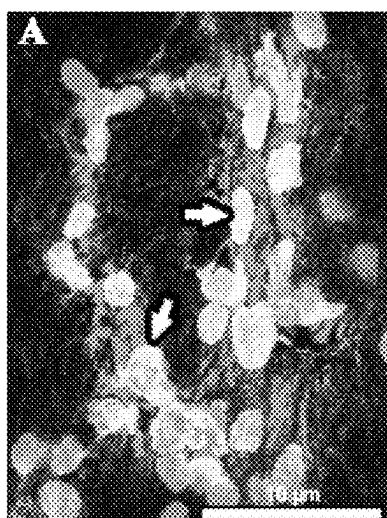


FIG. 8A

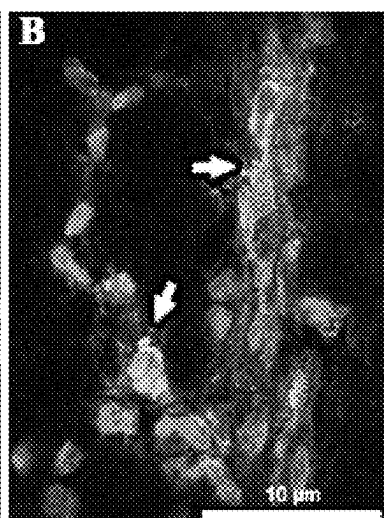


FIG. 8B

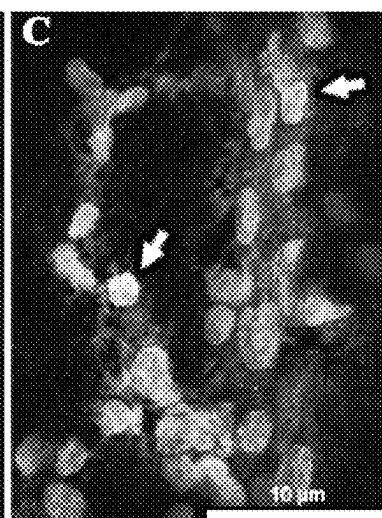


FIG. 8C

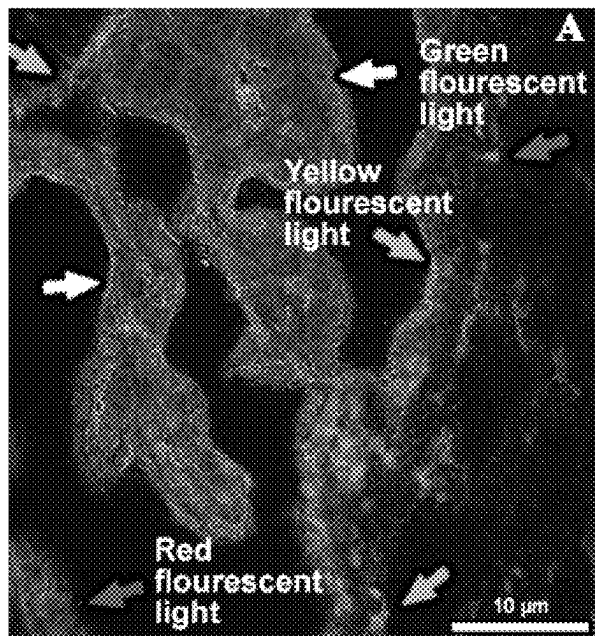


FIG. 9A

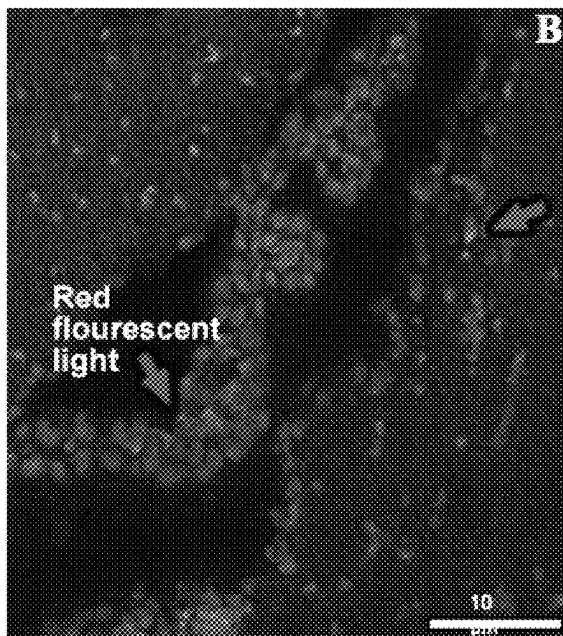


FIG. 9B

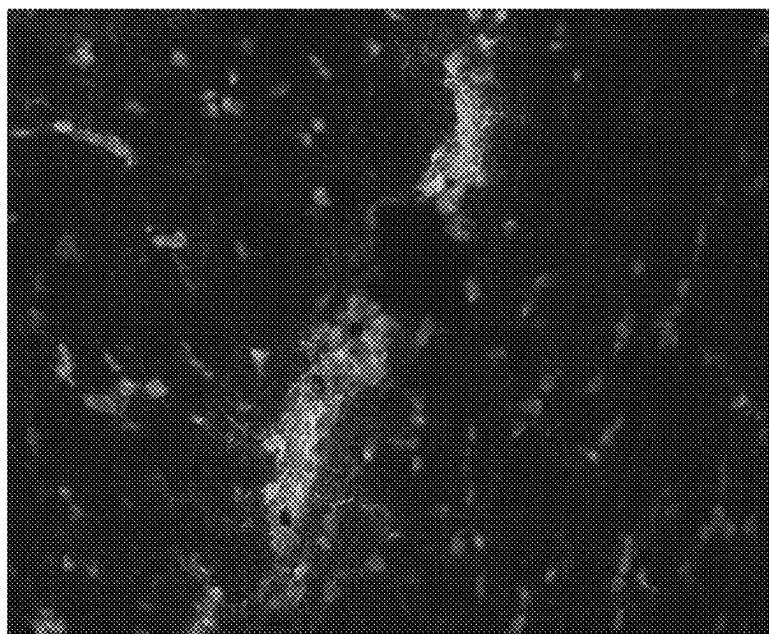


FIG. 9C

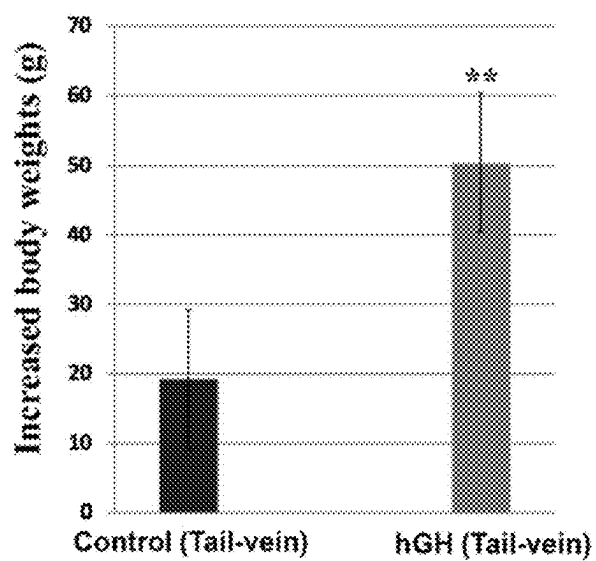


FIG. 10A

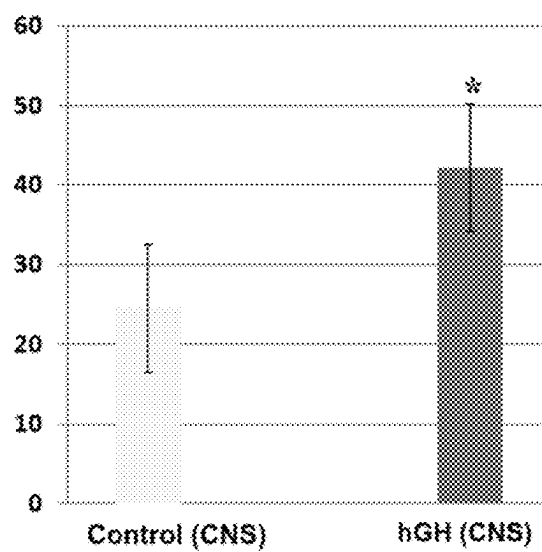


FIG. 10B

**VECTORS FOR DELIVERY OF HUMAN
GROWTH HORMONE GENE AND
MULTIPLE THERAPEUTIC GENES INTO
CENTRAL NERVOUS SYSTEM BY
CROSSING BLOOD BRAIN BARRIER**

**CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] This application claims the priority of U.S. Provisional Application No. 63/127,882, filed on Dec. 18, 2020, the content of which is incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure generally relates to the field of gene delivery technology and, more particularly, relates to vectors for delivery of human growth hormone gene and multiple therapeutic genes into a central nervous system (CNS) by crossing a blood brain barrier.

BACKGROUND

[0003] Growth hormone (GH) replacement therapy may improve mental disabilities in patients with adult GH deficiency and childhood-onset GH deficiency. In addition to normalizing growth and metabolism in adult patients with GH deficiency, the hormone therapy may enhance energy, motivation, and well-being, and may also improve memory and cognitive capacity. Neurons are considered to play an important role on brain memory and cognitive function. GH receptors for hormone targeting are present in the brain or central nervous system of human and animals. GH is known to mediate its effects through insulin-like growth factor-1 (IGF-1), and brain receptors for this mediator have also been demonstrated in areas associated with cognitive behaviors.

[0004] GH replacement therapy in patients who are suffered from GH deficiency (GHD) may reduce cognitive problem, improve memory, increase overall life quality, and may also show neuroprotection, learning and memory improvement, and promote growth in the pre-clinical study on animal models. Gene delivery of GH or related therapeutic proteins in target cells by vectors may be the most efficient treatment for various conditions.

[0005] Although human growth hormone therapy (hGH) is safe and effective under the range of applications approved by FDA (food and drug administration), it may like other protein therapies in long-term applications that may trigger immunoreactions due to different ages or populations. It has been reported that, for the treatment of hGH on 8 GHD children with ages from 1 to 17 for 16 months, B cells percentage was transiently decreased to subnormal levels in 7 of 8 patients. T helper/suppressor ratios were decreased in all patients to subnormal values in 7 of 8 patients; and mitogen responses were decreased to below normal in all patients. Generally, the immune response to protein-based therapeutics involves T-cells, which may help activate B cells to produce antibodies including those blocking protein therapeutics. The immune response may happen if the natural protein made by the body is defective to some extent. There is a need to develop a new construction for a specific delivery system to minimize above-mentioned side effects.

BRIEF SUMMARY OF THE DISCLOSURE

[0006] One aspect of the present disclosure provides expression vector for amplified expression in mammalian cells. The vector includes one or more inserted foreign DNA sequences, including human growth hormone (hGH) and a plurality of therapeutic genes, where the one or more inserted foreign DNA sequences are fused with intracellular or extracellular secretion sequences; a promoter, configured to drive expression of the one or more inserted foreign DNA sequences; an insulator, configured for stable and safety expression of the one or more inserted foreign DNA sequences; a marker gene, configured to encoding one or more corresponding proteins to indicate the expression of the one or more inserted foreign DNA sequences; and a regulator element, configured to control expression of the hGH and the plurality of therapeutic genes under a controllable condition.

[0007] Another aspect of the present disclosure provides a method for generating an expression vector for amplified expression in mammalian cells. The method includes fusing one or more inserted foreign DNA sequences with intracellular or extracellular secretion sequences, the one or more inserted foreign DNA sequences including human growth hormone (hGH) and a plurality of therapeutic genes; configuring a promoter to drive expression of the one or more inserted foreign DNA sequences; configuring an insulator for stable and safety expression of the one or more inserted foreign DNA sequences; configuring a marker gene to encoding one or more corresponding proteins to indicate the expression of the one or more inserted foreign DNA sequences; and configuring a regulator element control expression of the hGH and the plurality of therapeutic genes under a controllable condition.

[0008] Other aspects of the present disclosure can be understood by those skilled in the art in light of the description, the claims, and the drawings of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The following drawings are merely examples for illustrative purposes according to various disclosed embodiments and are not intended to limit the scope of the present disclosure.

[0010] FIG. 1 depicts an exemplary multiple gene expression vector according to various disclosed embodiments of the present disclosure;

[0011] FIG. 2A depicts an exemplary fluorescent image of in vitro transduction of brain choroid plexus cells by recombinant (adenoviral associate virus) AAVs according to various disclosed embodiments of the present disclosure;

[0012] FIG. 2B depicts an exemplary fluorescent image of in vitro transduction of brain choroid plexus cells infected by AAV vectors without any included transgenes (as control) according to various disclosed embodiments of the present disclosure;

[0013] FIG. 3A depicts an exemplary fluorescent image of intra-hippocampal injection of recombinant AAV vectors into the hippocampus of a rat brain according to various disclosed embodiments of the present disclosure;

[0014] FIG. 3B depicts an exemplary fluorescent image of intra-hippocampal injection of AAV vectors without any transgenes (as control) into the hippocampus of a rat brain according to various disclosed embodiments of the present disclosure;

[0015] FIG. 3C depicts an enlarge view of a boxed region (shows various CNS cells transduced by the transgenes from recombinant AAVs delivery) in FIG. 3A;

[0016] FIG. 4A depicts an exemplary fluorescent image from the expression of transgenes (fluorescent light) in CNS tissues after tail-vein injection of recombinant AAV vectors encoding transgenes of GFP and/or human growth hormone hGH in animal models according to various disclosed embodiments of the present disclosure;

[0017] FIG. 4B depicts an exemplary fluorescent image in a control animal model after tail-vein injection of recombinant AAV vectors without any transgenes according to various disclosed embodiments of the present disclosure;

[0018] FIG. 5A depicts an exemplary quantitative-polymerase chain reaction (qPCR) detection result of mRNA expressional levels of human growth hormone hGH after the tail-vein injection according to various disclosed embodiments of the present disclosure;

[0019] FIG. 5B depicts an exemplary quantitative-PCR (qPCR) detection result of mRNA expressional levels of human growth hormone hGH after intra-hippocampal injection according to various disclosed embodiments of the present disclosure;

[0020] FIG. 6A depicts an exemplary quantitative-PCR (qPCR) detection result of mRNA expressional levels of rat IGF-1 after the tail-vein injection according to various disclosed embodiments of the present disclosure;

[0021] FIG. 6B depicts an exemplary quantitative-PCR (qPCR) detection result of mRNA expressional levels of rat IGF-1 after intra-hippocampal injection according to various disclosed embodiments of the present disclosure;

[0022] FIG. 7A depicts an exemplary fluorescent image of neurons detected by containing a specific anti-NeuN antibody with IgG conjugated to TRITC (A594/red fluorescent light, 20×), where the slide was sectioned from CNS samples after the tail-vein injection, according to various disclosed embodiments of the present disclosure;

[0023] FIG. 7B depicts an exemplary fluorescent image for co-expression of hGH in the neurons by using a specific anti-human growth hormone antibody with IgG conjugated to FITC (A488/green fluorescent light, 20×), where the slide was sectioned from CNS samples after the tail-vein injection, according to various disclosed embodiments of the present disclosure;

[0024] FIG. 8A depicts an exemplary fluorescent image of neurons for a specific neural biomarker protein which was expressed in neurons of the CNS and detected by anti-NeuN antibody (white arrows indicate red fluorescent light), where the slide was sectioned from the CNS samples after the tail-vein injection, according to various disclosed embodiments of the present disclosure;

[0025] FIG. 8B depicts an exemplary fluorescent image for the expression of hGH proteins in the cytoplasm of the neurons, detected by a specific anti-human growth hormone antibody (white arrows indicate green fluorescent staining/light), where the slide was sectioned from the CNS samples after the tail-vein injection, according to various disclosed embodiments of the present disclosure;

[0026] FIG. 8C depicts an exemplary overlapped view of FIG. 8A and FIG. 8B;

[0027] FIG. 9A depicts an exemplary fluorescent image for the expression of hGH proteins detected by a specific antibody (green fluorescent staining/light) in CNS choroid plexus, neurons identified by a specific anti-NeuN antibody

(red fluorescent light), and overlapped co-localizations (yellow fluorescent light) of both hGH proteins and neurons in a same view of the tissues with the delivery of recombinant AAV vectors after tail-vein injection according to various disclosed embodiments of the present disclosure;

[0028] FIG. 9B depicts an exemplary fluorescent image for a control group without hGH protein expression according to various disclosed embodiments of the present disclosure;

[0029] FIG. 9C depicts an exemplary fluorescent image for the expression of hGH proteins in another section of CNS choroid plexus tissue with the delivery of recombinant AAV vectors after tail-vein injection according to various disclosed embodiments of the present disclosure;

[0030] FIG. 10A depicts effect of hGH treatment in tail-vein injection on body weights according to various disclosed embodiments of the present disclosure; and

[0031] FIG. 10B depicts effect of hGH treatment in CNS injection on body weights according to various disclosed embodiments of the present disclosure.

DETAILED DESCRIPTION

[0032] References may be made in detail to exemplary embodiments of the disclosure, which may be illustrated in the accompanying drawings. Wherever possible, same reference numbers may be used throughout the accompanying drawings to refer to same or similar parts.

[0033] A cloning vector, widely used in molecular biology, is a small piece of DNA molecule, which can be inserted with a foreign DNA fragment and used as a vehicle to transfer foreign genetic material into another cell. Insertion of the foreign DNA fragment into the cloning vector may be normally carried out by digesting both the vector and the foreign DNA with a restriction enzyme and ligating the fragments digested by the restriction enzyme together. The vector may be used for controlled expression of a particular gene with specific promoter sequences such as human neuron synapsin, neuron enolase (NSE), or neuron type-specific expression (CaM kinase II) to drive transcription of the transgene cloned in the vector.

[0034] Once the vector is inside a cell, the protein that is encoded by the transgene may be produced by cellular transcription and translation. After expression of the gene product, the resulted protein of interest may need to be purified and isolated from other proteins of the host cell. To facilitate the process, the cloned gene may normally a tag sequence, such as attached histidine (His) tag. In addition, GFP (green fluorescent protein) or other human antigen sequences may often be used as markers to follow the expression process for selection. In the cells where the tagged gene is expressed, the GFP may also be produced, and those cells may be observed under fluorescence microscopy and isolated by FACS (fluorescence-activated cell sorting).

[0035] However, overcoming the difficulty of delivering therapeutic agents to specific regions of the brain presents a major challenge to the treatment of most brain disorders. In brain's neuroprotective role, the blood brain barrier (BBB) may function to hinder the delivery of various potentially important diagnostic and therapeutic agents to the brain. Therapeutic molecules and antibodies that might otherwise be effective in diagnosis and therapy do not cross the BBB in effective amounts. The blood cerebrospinal fluid barrier, which is also a function of the choroidal cells of the choroid

plexus and forms the blood retinal barrier, may be considered a part of the whole realm of those barriers. For most CNS therapeutics on the market, less than 0.2% of the peripheral dose may be taken up by the brain. For example, only about 0.02% of the peripherally administered dose of morphine may enter the brain in order to produce sufficient analgesia via the CNS, but the remaining dosage in peripheral system may be more than 200 times higher than that in the brain.

[0036] According to various embodiments in the specification, an objective of the present disclosure is to deliver recombinant GH gene and/or other proteins (e.g., peptides) by a vector via crossing the BBB into neurons of the CNS to elicit therapeutic effect on patients who are suffered from neurodegenerative disorders such as intrauterine growth restriction, Alzheimer's disease, Parkinson disease, multiple sclerosis, Gaucher type-II or III disorders, neuropathic pain, spinal-cord injured, brain tumors, and the like, thereby promoting new neuron growth to recover biological functions of damaged neurons in clinical applications. Heterogenous genes or proteins used in the present disclosure may include, but not be limited to, IGF-1 (insulin growth factor-1), insulin, rat growth hormone, glucocerebrosidase, brain-derived neurotrophic factor (BDNF), β -endorphin, HSV-1 (herpes simplex virus), anti-glioma, ApoE4, prenilin 1 and/or 2, myelin oligodendrocyte glycoprotein (MOG), crry, gata4, glutamate decarboxylase 65, interleukin-10, chondroitinase ABC, P/NK-1R, human interleukin-12 (hIL-12), stem cell transcription factors for iPSC including October 3/4, Sox2, Klf1-5, Nanog, LIN28 and Glis1, and direct reprogramming genes: dopaminergic neurons reprogramming genes including ASC11, LMX1A, MYT1L and BRN2; direct glutamatergic, GABAergic neuronal reprogramming including ASC11, MYT1L, BRN2 and NeuroD1; direct retinal pigment epithelium reprogramming including PAX6, RAX, CRX, MITF-A, OTX2, NRL and KLF4/2; direct spinal motor neurons reprogramming including NGN2, SOX11, ISL1 and LHX3; anti-Tau, anti- β -amyloid antibodies, nanobodies (VHH), and/or any other suitable genes or proteins.

[0037] The construction of the vector encoding with hGH herein is able to cross the BBB, which may be further developed as a universal delivery system to carry any other suitable gene products to target the diseases in the CNS. The carriers may be viral gene vectors such as AAV and HIV, fusion protein or activated domain of polypeptides or peptides because the human growth (hGH) receptor is found in the human choroid plexus, which may indicate that hGH may enter the CNS via binding to its receptor on the choroid plexus tissue. The binding domain of such sequence may be selected and related sequences with capability to bind on the receptors of the BBB and may be inserted or constructed on the surfaces of the envelope proteins or capsulate proteins of the vector particles (shown in Table 1). Table 1 lists the sequences to be constructed on the surfaces of the envelopes and/or capsules of vector particles in the present disclosure. In addition, expression or production of fusion protein with capability to cross the blood brain barrier may be designed and realized in the vector construct using the vector system, which is potentially useful for the treatment of patients with CNS disorders and other diseases. The one or more inserted foreign DNA sequences may be fused with fusion sequences, which may be intracellular or extracellular secretion sequences. For example, the fusion sequences may

include ROIKIWFRNR, YGRKKRRORRR, and/or LLNFDLLKILLAGDWESNPCP, which may be fused with hGH and other therapeutic genes in the rAAV vector.

TABLE 1

The sequences to be constructed on the surfaces of the envelopes and/or capsules of vector particles in the present disclosure	Sequence names:
MFPTIPLSRLFDNAMLRAHRLHQLAF DTYQEFEEAYIPKEQKYSFLQNPQTS LCFSES	Human growth hormone receptor (hGHR) binding sequence 1
RTGQIFKQTYSKFDTNHNDALLKN YGLLYCFRKMDKVETFLRIVQCRSV EGSCGF	hGHR binding sequence 2
IPTPSNRETQQKSNLELLRLISLLIQS VYGASDSNVYDLLKDLEEGIQTLMGR LEDGSP	hGHR binding sequence 3
MDMRTPAQFLGILLWFPGLKCDIKM TQSPSSMYASLGERVFTFCKASQDIN NYLCWFQKPKGSKPTLIYRANRLVD GVPSRPSGSGSQDYSLTISSLEYED MGIYYCLQYDEFFPYTFGGGKLEIKR ADAA	The sequences designed to bind on the human transferrin receptor in the BBB
MAAAGQLCLLYLSAGLLSRLGAANL DTREDNVIRKYGDPGSLFGFSLAMHW QLQPEDKRLLLVGAPRAEALPLQRAN RTGGLYSCDI TARGPCTRIEFNDAD PTSESKEDQWMGVTVQSQGPGKVV CAHRYEKRQHVNTKQESRDIFGRCYV LSQNLRIEDMDGGDWSFCDGRLRGH EKGSCQQGVAATFTKDF	The antibody sequences designed to bind on the Integrin $\alpha 6 \beta 4$
VQLQQSGAELVKPGASVKLSCTASGF NIKDTYMHVVKQRPEQGLEWIGRIDP ASGDTKYDPKFQVRVITADTSTDTA YMELSSLSRSEDVAVYCATGMWVSTG YALDFWGQGTLLVTVS	The sequences designed to bind on Integrins in the BBB
EVQLQQSGPELVKPGDSVKMSCKASG YTFDYYMDWVKQSHGKSLIEWIGYIY PNNGGTSYNQKFKGKATLTVDKSSST AYMELHSLTSEDSAVVYCAR	The sequences designed to bind on laminin in the BBB

[0038] Stem cell therapy may only work desirably for long-term clinical treatment of genetic modifications with encoding particularly biological functions. The stem cells transduced by the vector system encoding growth hormone gene and/or other genes (i.e., insulin, IGF-1, reprogramming factors) may be induced and differentiated into specific cell types such as the neurons being capable to secrete hGH and/or express desirable therapeutic proteins which may be subsequently transplanted in the CNS of neurodegenerative patients, the islet cells secreting insulin implanted into pancreatic tissues for diabetic patients, and the bone marrow stem cells with expressing normal enzyme of glucocerebrosidase infused to the patients with Gaucher disease and the like.

[0039] A brain tumor is an intracranial solid neoplasm within the brain or the central spinal canal. Any brain tumor is inherently serious and life-threatening because of its invasive and infiltrative character in the limited space of the

intracranial cavity. The treatment of such disease may be limited due to the BBB, and the vector system in the present disclosure may provide an alternative and useful platform to deliver therapeutic genes or proteins in the target tissues or cells in the CNS crossing the BBB.

[0040] The viral gene vector may be used as a useful investigative tool. For example, for an animal model of Sjorgren syndrome, an autoimmune disorder was generated by using the AAV vector encoding a gene of BMP6 (bone morphogenetic protein 6), which is up-regulated in both patients and Aec1/Aec2 knock-out mice. The AAV-BMP6 vector was delivered into the submandibular glands by retrograde instillation. For another example, the delivery of a gene encoding a beta-amyloid, a precursor protein leading neurofibrillary tangles and plaques found in Alzheimer's brain, into the cortex or hippocampus of animal brain by the vector may generate a rapid animal model of Alzheimer's disease. Such approach may be extremely helpful for rapid generation of an acute animal (rat) model, which may be a desirable pre-clinical model for behavior testing required in pharmaceutical industry.

[0041] This viral gene vector may be designed to encode and/or to express the "biological tracers" to label tissues or cells, such as (AAV) viral labeled neurons in vitro and in vivo, which may be combined with laser-captured microscope to select a single cell (neuron) or sample to perform microarray study or may monitor AMPA (α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid) receptor (i.e. expressing GluR1-GFP fusion) trafficking during neuronal synaptic plasticity to understand the mechanism of learning, memory, and drug addiction.

[0042] In various embodiments of the present disclosure, a gene-based therapy may be developed to avoid potential immune response from a therapeutic protein in the delivery system. The vector delivery system may be genetically engineered under multiple regulator elements such as the capsule or envelope of the viral gene-vector with specific binding to target the tissues or cells where the therapeutic protein is expressed under a specific promoter such as hGH in the neurons. In one embodiment, the regulator elements may include (1) Tetracycline response element (TRE) including the sequences of "TCCCTATCAGTGA-TAGAGA" to be included into the vector, for example placed it in front of the promoter, in order to turn on or turn off the expression of transgenes or therapeutic genes under the condition in the present of inducer, such as by doxycycline reagent or drug; and/or (2) post-transcriptional regulatory element (PRE)-derived from the woodchuck hepatitis virus (WHV) into the 3' untranslated region of AAV or lentiviral gene transfer vectors to enhance both titer and transgene expression. In embodiments, the regulator elements may include sequences of "ttgtcaggc aacgtggcgt ggtgtgcact gtgtttgctg acgcaacccc cactggttgg ggcattgccca ccactgtca gctccttcc gggactttcg ctttccccct cctattgcc acggcggaac tcactgcgcg ctgccttgcc cgctgctgga caggggctcg gctgtgggc actgacaatt ccgtggtgtt gtcggggaag ctgacgtct ttc-catggct gctegcctgt gttgccact ggattctgcg c". Therefore, patients may be treated with the therapeutic protein which is less likely to cause immune response and other side effects. For example, a multiple and regulated rAAV and/or lentiviral gene-vector system may be generated.

[0043] In various embodiments of the present disclosure, a multiple viral gene vector may be constructed, where the gene vector may include mammalian cell promoters or cell

specific promoters, inserted nucleotide sequences encoding hGH and therapeutic genes, and biomarkers for treating the CNS diseases.

[0044] In various embodiments of the present disclosure, a viral gene vector, including a neuronal promoter or other specific promoters that function in neuron cells and in specific tissues or specific cell types, may be constructed.

[0045] In various embodiments of the present disclosure, a method for delivery of the gene encoding a therapeutic protein and hGH into the CNS by crossing the BBB may be provided.

[0046] In various embodiments of the present disclosure, a gene-based therapy may be provided to cure neurological disorders, neuropathic pain by crossing the BBB, by inserting therapeutic gene sequences into the vector, hGH, glucocerebrosidase, beat-endorphin, antisense, shRNA, mRNA, CRISPR/Cas9, iPSC stem cell transcription factors, and tissue direct reprogramming factors.

[0047] In various embodiments of the present disclosure, a combination technique of genetic engineering stem cells transduced by the vector encoding hGH may be provided to produce long-term and stable therapeutic effect on patients with the spinal-cord injured, skin burned, diabetes, and other related disease, with the combination with cell therapy or cell transplantation.

[0048] Various embodiments of the present disclosure provide a delivery system for carrying the therapeutic genes with direct iPSC (induced pluripotent stem cells) functions such as Sox2, Oct4, Nanog, Lin28, Klf2 and other related stem cell(s). Only one gene and/or a combination of multiple same or different iPSC genes may be used in the present disclosure, which may not be limited according to various embodiments of the present disclosure. For example, one gene such as SOX2 in a vector system may be used for treatment, but also two or more genes, such as SOX2 and Oct4 or SOX2, Oct4, and Naong cloned in a vector system, may be used for treatment. The degeneration of neuronal or CNS cells may be converted stem cell-like cells and then subsequently differentiated back to original neurons and other CNS cells with normal functionals in the microenvironment within the CNS system and/or after crossing the BBB.

[0049] Various embodiments of the present disclosure provide a delivery system of carry the therapeutic genes with direct tissue specific reprogramming functions such as ASC11, LMX1A, MYT1L, BRN2, NeuroD1, PAX6, RAX, CRX, MITF-A, OTX2, NRL, NGN2, SOX11, ISL1, LHX3 and other related direct reprogramming factors and/or genes. Only one gene and/or combination of multiple same or different direct tissue specific reprogramming factors or genes may be used in the present disclosure, which may not be limited according to various embodiments of the present disclosure. For example, one gene such as ASC11 in a vector system may be used for treatment, but also two or more genes, such as ASC11 and MYT1L, or MYT1L, BRN2, and NeuroD1 cloned in a vector system, may be used for treatment. The degeneration of neuronal or CNS cells may be converted to be reprogrammed and differentiated back to original neurons and other CNS cells with normal functionals in the microenvironment within the CNS system and/or after crossing the BBB.

[0050] Various embodiments of the present disclosure provide a delivery system of carry the therapeutic genes with any of combinations from the direct reprogramming factors

and iPSC stem cell transcriptional factors; and/or combination of multiple same or different factors or genes. The degeneration of neuronal or CNS cells may be converted to be reprogrammed and turned into stem cell-like cells by iPSC and then subsequently differentiated back to original neurons and other CNS cells with normal functionals in the microenvironment within the CNS system and/or after crossing the BBB.

[0051] In various embodiments of the present disclosure, a delivery system, carrying specific anti-cancer genes, antibodies of full or single-chain variable fragment (scFv), nanobody of VHH (variable heavy homodimers), proteins, and reagents to treat the CNS diseases including tumors or cancers by crossing the BBB, may be provided.

[0052] Various embodiments of the present disclosure may provide a method for generating an investigative tool as “biological tracer” to study or recognize the signal interaction among cell-types and to directly deliver transgene to a particular region to produce a rapid rodent animal model (rat) for pharmaceutical research and industry.

[0053] Various embodiments of the present disclosure provide a vector for delivery of human growth hormone genes and multiple therapeutic genes into a central nervous system by crossing a blood brain barrier. The vector may be a viral vector, and also be a non-viral vector such as lipid nanoparticles (LNP), which may not be limited according to various embodiments of the present disclosure. The exemplary vector is described in detail according to the present disclosure hereinafter.

[0054] FIG. 1 depicts an exemplary multiple gene expression vector according to various disclosed embodiments of the present disclosure.

[0055] Referring to FIG. 1, the AAV multiple gene vector may be constructed with a human specific neuronal promoter of h-Synapsin followed by inserted nucleotide sequences of a cDNA encoding hGH which is linked to a growth hormone genome sequence named as “GS” and is also linked to a 2A sequence or other cleavable sequence for expression of a marker gene of GFP or Laz. In addition of such new AAV gene expression cassette, an adenovirus helper and a AAVRH 10 of Rep-Cap may be co-transfected into 293T cells to generate recombinant AAVs, which may express the hGH protein specified in the CNS neurons by crossing the BBB.

[0056] In one embodiment, additional functional sequences may be included in the present disclosure. For example, insulators such as chromatin insulators from mammalian cells may be constructed in the vectors for stable and safety expression of therapeutic genes without toxicity due to the possible integration of the vectors. Relevant sequences of the insulators may be summarized in the Table 2. Table 2 lists exemplary insulators DNA sequences used in the present disclosure. The insulator used herein may include a part or whole of the first chromatin insulator, a part or whole of the second chromatin insulator, and/or a part or whole of the third chromatin insulator. Other variant sequences with the same function as insulators, including AAV-insulator-h-Synapsin (promoter)-therapeutic gene (hGH)-insulator, may not be limited according to various embodiments of the present disclosure.

TABLE 2

Chromatin insulators DNA sequences used in the present disclosure	Names of insulators
GAGCTCACGGGGACAGCCCCCCCCAAAGCCCCAGGGA TGTAATTACGTCCCTCCCCCGCTAGGGGCGAGCAGCGAG CCGCCCCGGGGCTCCGCTCCGGTCCGGCGCTCCCCCGCA TCCCGAGCCGGCAGCGTGCGGGGACAGCCCCGGGCACGG GGAAGGTGGCAGGGATCGCTTTCCTGAAACGCTTCTC GCTGCTCTTTGAGCCTGCAGACACCTGGGGGGATACGGG GAAAAAGCTTTAGGCTGAAAGAGAGATTTAGAAATGACAG GCGCGCTTGGCCATACATCGATACGGTACCGAGTTGGCG CGCCTGGGAGCTCACGGGGACAGCCCCCCCCAAAGCCC CCAGGGATGTAATTACGTCCCTCCCCCGCTAGGGGGCAG CAGCGAGCCCGGGGGCTCCGCTCCGGTCCGGCGCTCC CCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCG GGCACGGGGAAGGTGGCAGGGATCGCTTTCCTCTGAAC GCTTCTCGCTGCTCTTTGAGCCTGCAGACACCTGGGGGG ATACGGGAAAAAGCTT	First chromatin insulator
CACAGGTGCTAATATTTCATTTATAGCTCTAAATCAGTGG CCAAAAGCGTGGCCCCGACAAAAGGCTTCTATGGGCTTGA TTGGAGTAGGCACAGCATCTGAAGTCTATCAAAGTGCTAC AATTTTACATCGTTTAAGACCAGATGGGCAGGAAGTCAAC ATCCAACCTATGATTACTTTTCATTCCCATTAATTTGTAGG GGAGAGATTTGTTACAGCAATGGGGTACAGAAATCACCAT TCCCTCTGCAATTTACAGTCTGAGAGCCAAAGAATGATG ACCAAAATGGGATATATTCTGGGAAAGGATTAGGAAAAA AATGAGGACGGAATAACCCAGCCGATTGAGGTCTCAGTTA AATTTGACCGAAGAGGAATTTGATATCCTTCTTAGGTGCG GCCACTGTTGAGCCTCAAAGCCCATTCATTAACCGGA AAACTCCAAACCTGTTTGGGTTGATCAGTGGCCGCTTCC TCAACAAAACTGGAGGCATTACATTCATTGGCAAAAGAA CAATTAGACAAAGGACATATTGAACCATCTTTTTCACCAT GGAACGCCCCAGTTTTTGTAAATCAGAAAAAATTCGCTAG ATGGCGCATATTGACCGACTTATGAGCGGTTAACGAGCC ATTCAACCGATGGGAGCTCTCCAACAGGGTTGCCATCTC CGGCCATGATCCCAAAGAAATGGCCGATGGTAATAATTGA TCTAAAAGATTGTGTTTTTACTATTCCTTTGGCTCCTCA	Second chromatin insulator

TABLE 2-continued

Chromatin insulators DNA sequences used in the present disclosure	Names of insulators
GGATTTTGAGAAATTGCAATTTACAATACCTGCCATGAAT AACAAAGGAACCACTACTATATATCGATGGAAAGTTGTAC CCCAAGGAATGTAAATAGTCAAACCTTTTGTGGAAAGGT TATTGAGCTTGTGAGGATCAGTTTCTGATTGCTATATT ATTGATTATGTTGATGACATTTATGTGCTGCTGCAAGCA GGGACAGACGGATTGAGTGTGTTGCTAGGCTACAAGAAGT GGTGGGACTTACAGGTCTCGCTATAGCACCAGATAAAATC CATACTACACCTATCATTACCTGGGAATGAG	
GAAGGTGA TGCAGTCGAA GCCATTGTGG AGGAGTCCGA AACTTTTATTAAGGAAAGG AGAGAAAGAC TTACCAGAGA CGCCGGGAAG GGGGCCAGGA AGAAGATGCC TGCCACTTAC CCCAGAACCAGACGGATGGG GGTGAGGTGG TCCAGGATGT CAACAGCAGT GTACAGATGG TGATGATGGA ACAGCTGGAC CCCACCCCTTTC TTCAGATGAA GACTGAAGTA ATGGAGGGCA CAGTGGCTCC AGAAGCAGAG GCTGCTGTGG ACGATACCCA GATTATACT TTACAGGTTG TAAATATGGA GGAACAGCCC ATAAACATAG GAGAACTTCA GCTTGTTCAT GTACCTGTTC CTGTGACTGT ACCTGTTGCT ACCACTTCAG TAGAAGAACT TCAGGGGGCT TATGAAATG AAGTGTCTAA AGAGGGCCTT GCGGAAAGTG AACCCATGAT ATGCCACACC CTACCTTTGC CTGAAGGGTT TCAGGTGGTT AAAGTGGGGG CCAATGGAGA GGTGGAGACA CTAGAACAAG GGGAACTTCC ACCCCAGGAA GATCCTAGTT GGCAAAAAGA CCCAGACTAT CAGCCACCAG CCAAAAAAGC AAAGAAAACCAAAAAGAGCA AACTGCGTTA TACAGAGGAG GGCAAAGATG TAGATGTGTC TGTCTACGAT TTTGAGGAAG AACAGCAGGA GGGTCTGCT TCAGAGGTGA ATGCAGAGAA AGTGGTTGGT AATATGAAGC CTCCAAAGCC AACAAAAATT AAAAAGAAAG GTGTAAAGAA GACATTCCAG TGTGAGCTTT GCAGTTACAC GTGTCCACGG CGTTCAAATT TGGATCGTCA CATGAAAAGC CACACTGATG AGAGACCACA CAAGTGCCAT CTCGTGGCA GGGCATTCAG AACAGTCACC CTCCTGAGGA ATCACCTTAA CACACACACA GGTACTCGTC CTCACAAGTG CCCAGACTGC GACATGGCCT TTGTGACCAG TGGAGAATTG GTTCGGCATC GTCGTTACAA ACACACCCAC GAGAAGCCAT TCAAGTGTTT CATGTGCGAT TACGCCAGTG TAGAAGTCAG CAAATTAATA CGTCACATTC GCTCTCATAC TGGAGAGCGT CCGTTTCAGT GCAGTTTGTG CAGTTATGCC AGCAGGGACA CATAAAGCT GAAAAGGCACATGAGAACCC ATTCAGGGGA AAAGCCTTAT GAATGTTATA TTTGTCATGC TCGGTTTACC CAAAGTGGTA CCATGAAGAT GCACATTTTA CAGAAGCACA CAGAAAATGT GGGCAAATTT CACTGTCCCC ACTGTGACAC AGTCATAGCC CGAAAAAGTG ATTTGGGTGT CCACTTGCGA AAGCAGCATT CCTATATTGA GCAAGGCAAG AAATGCCGTT ACTGTGATGC TGTGTTTCAT GAGCGCTATG CCCTCATCCA GCATCAGAAG TCACACAAGA ATGAGAAGCG CTTTAAGTGT GACCAAGTGT ATTACGCTTG TAGACAGGAG AGGCACATGA TCATGCACAA GCGCACCCAC ACCGGGGAGA AGCCTTACGC CTGCAAGCCAC TCGGATAAGA CCTTCCGCCA GAAGCAGCTT CTCGACATGC ACTTCAAGCG CTATCACGAC CCCAACTTCG TCCCTGCGGC TTTTGTCTGT TCTAAGTGTG GGAAAACATT TACACGTCGG AATACCATGG CAAGCATGCG TGATAATTGT GCTGGCCAG ATGGCGTAGA GGGGGAAT GGAGGAGAAA CGAAGAAGAG TAAACGTGGA AGAAAAAGAA AGATGCGCTC TAAGAAAGAA GATTCTCTG ACAGTGAAAA TGCTGAACCAGATCTGGACG ACAATGAGGA TGAGGAGGAG CCTGCCGTAG AAATTGAACC TGAGCCAGAGCCTCAGCCTG TGACCCAGC CCCACACCC GCCAAGAAGC GGAGAGGACG ACCCCTGGCAGAACCAAC AGCCCAAACA GAACCAAGCA ACAGCTATCA TTCAGGTTGA AGACCAGAAT ACAGGTGCAA TTGAGAACAT TATAGTTGAA GTAAAAAAG AGCCAGATGC TGAGCCCGCA GAGGGAGAGG AAGAGGAGGC CCAGCCAGCT GCCACAGATG CCCCAACGG AGACCTCACG CCCGAGATGA TCCTCAGCAT GATGGACCGG	Third chromatin insulator

[0057] In one embodiment, specific promoters for transducing the CNS cell types, such as h-Synapsin promoter and h-GFAP promoter, may also be included.

[0058] In one embodiment, the expression cassettes (EC) of the vector of AAV, or Lentiviral vector or LNP, may also include the following sequences for enhancing transgene expression: poly Y-N-C-A-G-G (splicing acceptor

sequences) at the 5' end of the EC, and G-G-G-U-R-A-G-U (splicing donor sequences) at the 3' end of the EC. For example, the expression cassette in the AAV vector: AAV-poly Y-N-C-A-G-G-insulator-h-Synapsin (promoter)-therapeutic genes (hGH-2A-NGN2-2A-SOX2)-insulator-G-G-G-U-R-A-G-U.

[0059] In one embodiment, in order to deliver recombinant AAV or lentiviral vectors into the CNS by crossing the BBB, envelope and capsule with genetic engineered polypeptides (as shown in Table 1) may target the receptors located on the BBB.

[0060] Exemplarily, referring to FIG. 1, the vector, designed on its application for delivery and production of the protein such as hGH in the CNS neurons in vivo and mammalian cells in vitro, may include a cytomegalovirus (CMV) promoter, a glial fibrillary acidic protein (GFAP) promoter for astrocyte in the CNS, modified nucleotide sequences, and a GFP gene. The vector may be digested by BamH-I and Hind III at 37° C. for 1.5-2 hours; the sequence of insert-1 of hGH gene may be cloned into the digested AAV-2 vector; and the sequence of insert-2 may be isolated from a pUC12 vector by digestion of Hind III and Cla I.

[0061] The sequence of cDNA of hGH may be generated by polymerase chain reaction (PCR) using the primers designed as shown in the following. Designed PCR primers for cloning human growth hormone gene into the vector with human synapsin neuron specific promoter may be illustrated herein, where the PCR primers may include a forward primer (A) and a reverse primer (B).

A) Forward primer (sense):
5'-CTG CCC CAA GTC GCA GCC ATG GCT ACA GGC TCC CGG ACG-3'
Vector (hSyn) Nco-I hGH(start sequence)

B) Reverse primer (antisense):
5'-G CTT ATC ATG TCT GGC CAG CT AG CTA GAA GCC ACA GCT GCC-3'
Vector (hSyn-Lucia) Nhe-I hGH (end sequence)

[0062] Exemplarily, the PCR product fragment of hGH with primers containing partial sequences that are overlapped with the neuronal vector may be cloned as a new vector B containing cDNA of hGH linked to a gene of GFP or Laz, using a Cold Fusion Protocol (System Bioscience). The vector B may then be digested by Ase I and Hind III. The fragment containing two genes expression cassettes driven by the neuronal promoter of hSynapsin may be subsequently cloned back to the region of the vector A by digestion of Not I and Cla I to generate a new vector named as GS vector.

[0063] FIG. 2A depicts an exemplary fluorescent image of in vitro transduction of brain choroid plexus cells by recombinant (adenoviral associate virus) AAVs according to various disclosed embodiments of the present disclosure; and FIG. 2B depicts an exemplary fluorescent image of in vitro transduction of brain choroid plexus cells infected by AAV vectors without any included transgenes (as the control) according to various disclosed embodiments of the present disclosure. In various embodiments, serotypes of AAV vectors may be, such as AAV1, AAV2, AAV3, AAV 4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV (Rh) 10, AAV11, AAV12, etc.

[0064] Referring to FIG. 2A, it may indicate newly prepared AAVs that carry both GFP and human growth hormone. Only 1.0 μ L of such viral vector with titer (concentration) of 10^{13} pfu/mL may be applied to sufficiently transduce the humane choroid plexus cells (4×10^4) in vitro.

[0065] The new multiple gene vector may be capable of delivering at least two proteins which include a therapeutic protein (i.e., hGH) expressing in mammalian cells and another protein as a biomarker (e.g., GFP) in the cells

transduced by such vector system, where the transduced cells may be selected and distinguished from un-transduced cells. In addition, these proteins may be expressed in specific cell-types and/or tissues, for example, neurons may be driven by a neuronal promoter.

[0066] According to various embodiments of the present disclosure, an expression vector for amplified expression in mammalian cells is provided. The vector includes one or more inserted foreign DNA sequences, including human growth hormone (hGH) and a plurality of therapeutic genes, where the one or more inserted foreign DNA sequences are fused with intracellular or extracellular secretion sequences; a promoter, configured to drive expression of the one or more inserted foreign DNA sequences; an insulator, configured for stable and safety expression of the one or more inserted foreign DNA sequences; a marker gene, configured to encoding one or more corresponding proteins to indicate the expression of the one or more inserted foreign DNA sequences; and a regulator element, configured to control expression of the hGH and the plurality of therapeutic genes under a controllable condition.

[0067] In one embodiment, the expression vector includes an adenoviral associate virus (AAV) vector and/or a lenti-

viral vector (e.g., including a human immunodeficiency virus (HIV) vector), and/or a non-viral vector (e.g., including a lipid nanoparticle (LNP) vector).

[0068] In one embodiment, the promoter includes a cytomegalovirus (CMV) promoter or an elongation factor (EF-1) promoter or a synthetic specific promoter.

[0069] In one embodiment, the marker gene includes a green fluorescent protein (GFP) gene and/or a non-toxic mammalian biomarker.

[0070] In one embodiment, the expression vector further includes a neuronal promoter, including human synapsin, configured to drive expression of transgenes in neurons in a central nervous system (CNS).

[0071] In one embodiment, the insulator includes one or more of a first chromatin insulator, a second chromatin insulator, and a third chromatin insulator.

[0072] The first chromatin insulator includes at least a part of a sequence of:

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GAGCTCACGGGACAGCCCCCCCCAAAGCCCCCAGGGA
TGTAATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAG
CCGCCCCGGGCTCCGCTCCGGTCCGGCGCTCCCCCGCA
TCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGCACGG
GGAAGGTGGCAGGGATCGCTTTCTCTGAACGCTTCTC
GCTGCTCTTTGAGCCTGCAGACACCTGGGGGATACGGG
GAAAAAGCTTTAGGCTGAAAGAGAGATTAGAAATGACAG
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GCGCGCCTGGCCATACATCGATACGGTACCGAGTTGGCG
 CGCCTGGGAGCTCACGGGGACAGCCCCCCCCAAAGCCC
 CCAGGGATGTAATTACGTCCCTCCCCCGCTAGGGGGCAG
 CAGCGAGCCGCGGGGCTCCGCTCCGGTCCGGCGCTCC
 CCCCGCATCCCCGAGCCGGCAGCGTGCGGGACAGCCCG
 GGCACGGGAAGGTGGCACGGGATCGCTTTCCTCTGAAC
 GCTTCTCGTCTCTTTGAGCCTGCAGACACCTGGGGGG
 ATACGGGAAAAAGCTT;

[0073] The second chromatin insulator includes at least a part of a sequence of:

CACAGGTGCTAATATTTCAATTATAGCTCTAAATCAGTGG
 CCAAAAGCGTGGCCCCGACAAAAGGCTTCTATGGGCTTGA
 TTGGAGTAGGCACAGCATCTGAAGTCTATCAAAGTGCTAC
 AATTTTACATCGTTTAAGACCAGATGGGCAGGAAGTCACC
 ATCCAACCTATGATTACTTTTCATTCCCATTAATTTGTAGG
 GGAGAGATTTGTTACAGCAATGGGTACAGAAATCACCAT
 TCCCTCTGCAATTTACAGTCTGAGAGCCAAAGAATGATG
 ACCAAAATGGGATATATTCCTGGGAAAGGATTAGGAAAA
 AATGAGGACGGAATAACCCAGCCGATTGAGGTCTCAGTTA

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AATTTGACCGAAGAGGAATTGGATATCCTTCTTAGGTGCG
 GCCACTGTTGAGCCTCCAAAGCCCATTGCATTAAAAACGGA
 AAACCCAAAACCTGTTTGGGTTGATCAGTGCCCGCTTCC
 TCAACAAAAACTGGAGGCATTACATTCATTGGCAAAAGAA
 CAATTAGACAAAGGACATATTGAACCATCTTTTTCACCAT
 GGAACGCCCCAGTTTTTGTAAATTCAGAAAAAATTCGCTAG
 ATGGCGCATATTGACCGACTTATGAGCGGTTAACGCAGCC
 ATTCAACCGATGGGAGCTCTCCAACAGGGTTGCCATCTC
 CGGCCATGATCCCAAAGAATGGCCGATGGTAATAATTGA
 TCTAAAAGATTGTGTTTTTACTATTTCCTTTGGCTCCTCA
 GGATTTTGAGAAATTTGCATTTACAATACCTGCCATGAAT
 AACAAGGAACAGCTACTATATATCGATGGAAAGTTGTAC
 CCCAAGGAATGTTAAATAGTCAAACCTTTTGTGGAAAGGT
 TATTCAGCTTGTGAGGATCAGTTTCTGATTGCTATATT
 ATTCAATTATGTTGATGACATTTTATGTGCTGCTGCAAGCA
 GGGACAGACGGATTGAGTGTGTTGTAGCTACAAGAAGT
 GGTGGGACTTACAGGTCTCGCTATAGCACCAGATAAAATC
 CATACTACCCCTATCATTACCTGGGAATGAG.

[0074] The third chromatin insulator includes at least a part of a sequence of:

GAAGGTGA TGCAGTCGAA GCCATTGTGG AGGAGTCCGA
 AACTTTTATTAAAGGAAAGG AGAGAAAGAC TTACCAGAGA CGCCGGGAAG
 GGGGCCAGGA AGAAGATGCC TGCCACTTAC CCCAGAACCAGACGGATGGG
 GGTGAGGTGG TCCAGGATGT CAACAGCAGT GTACAGATGG TGATGATGGA
 ACAGCTGGAC CCCACCCTTC TTCAGATGAA GACTGAAGTA ATGGAGGGCA
 CAGTGGCTCC AGAAGCAGAG GCTGCTGTGG ACGATACCCA GATTATAACT
 TTACAGGTTG TAAATATGGA GGAACAGCCC ATAAACATAG GAGAACCTCA
 GCTTGTTCAG GTACCTGTTC CTGTGACTGT ACCTGTTGCT ACCACTTCAG
 TAGAAGAACT TCAGGGGGCT TATGAAAATG AAGTGTCTAA AGAGGGCCTT
 GCGGAAAGTG AACCCATGAT ATGCCACACC CTACCTTTGC CTGAAGGGTT
 TCAGGTGGTT AAAGTGGGGG CCAATGGAGA GGTGGAGACA CTAGAACAAG
 GGGAACTTCC ACCCCAGGAA GATCCTAGTT GGCAAAAAGA CCCAGACTAT
 CAGCCACCAG CCAAAAAAAC AAAGAAAACCAAAAAGAGCA AACTGCCTTA
 TACAGAGGAG GGCAAGATG TAGATGTGTC TGTCTACGAT TTTGAGGAAG
 AACAGCAGGA GGGTCTGCT TCAGAGGTTA ATGCAGAGAA AGTGGTTGGT
 AATATGAAGC CTCCAAGCC AACAAAAATT AAAAAGAAAG GTGTAAGAA
 GACATTCCAG TGTGAGCTTT GCAGTTACAC GTGTCCACGG CGTTCAAATT
 TGGATCGTCA CATGAAAAGC CACACTGATG AGAGACCACA CAAGTGCCAT

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CTCTGTGGCA GGGCATTGAG AACAGTCACC CTCCTGAGGA ATCACCTTAA
CACACACACA GGTACTCGTC CTCACAAGTG CCCAGACTGC GACATGGCCT
TTGTGACCAG TGGAGAATTG GTTCGGCATC GTCGTTACAA ACACACCCAC
GAGAAGCCAT TCAAGTGTTT CATGTGCGAT TACGCCAGTG TAGAAGTCAG
CAAATTAAAA CGTCACATTC GCTCTCATAC TGGAGAGCGT CCGTTTCAGT
GCAGTTTGTG CAGTTATGCC AGCAGGGACA CATACAAGCT
GAAAAGGCACATGAGAACCC ATTCAGGGGA AAAGCCTTAT GAATGTTATA
TTTGTATGTC TCGGTTTACC CAAAGTGGTA CCATGAAGAT GCACATTTTA
CAGAAGCACA CAGAAAATGT GGCCAAATTT CACTGTCCCC ACTGTGACAC
AGTCATAGCC CGAAAAAGTG ATTTGGGTGT CCACTTGCGA AAGCAGCATT
CCTATATTGA GCAAGGCAAG AAATGCCGTT ACTGTGATGC TGTGTTTCAT
GAGCGCTATG CCCTCATCCA GCATCAGAAG TCACACAAGA ATGAGAAGCG
CTTTAAGTGT GACCAGTGTG ATTACGCTTG TAGACAGGAG AGGCACATGA
TCATGCACAA GCGCACCAC ACCGGGAGGA AGCCTTACGC CTGCAGCCAC
TGCATAAGA CCTTCCGCCA GAAGCAGCTT CTCGACATGC ACTTCAAGCG
CTATCACGAC CCCAACTTCG TCCCTGCGGC TTTTGTCTGT TCTAAGTGTG
GGAAAACATT TACACGTCGG AATACCATGG CAAGACATGC TGATAATTGT
GTGCGCCAG ATGGCGTAGA GGGGGAAAAT GGAGGAGAAA CGAAGAAGAG
TAAACGTGGA AGAAAAAGAA AGATGCGCTC TAAGAAAGAA GATTCTCTCG
ACAGTGAAAA TGCTGAACCAGATCTGGACG ACAATGAGGA TGAGGAGGAG
CCTGCGGTAG AAATTGAACC TGAGCCAGAGCCTCAGCCTG TGACCCACG
CCCACCACCC GCCAAGAAGC GGAGAGGACG ACCCCCTGGCAGAACCAACC
AGCCCAAACA GAACCAGCCA ACAGCTATCA TTCAGGTTGA AGACCAGAAT
ACAGGTGCAA TTGAGAACAT TATAGTTGAA GTAAAAAAG AGCCAGATGC
TGAGCCCGCA GAGGGAGAGG AAGAGGAGGC CCAGCCAGCT GCCACAGATG
CCCCAACGG AGACCTCACG CCCGAGATGA TCCTCAGCAT GATGGACCGG

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[0075] In one embodiment, a surface capsule of the expression vector is engineered by inserting a nucleotide sequence to bind and/or recognize one or more receptors on Choroid plexus and blood brain barrier (BBB) tissues.

[0076] In one embodiment, a surface capsule of the expression vector is engineered by inserting a single-chain fragment variable (scFv) antibody sequence to bind and/or recognize one or more receptors on Choroid plexus and blood brain barrier (BBB) tissues.

[0077] In one embodiment, a surface capsule of the expression vector is engineered by inserting a ligand sequence derived from the hGH to bind and/or recognize one or more hGH receptors on Choroid plexus and blood brain barrier (BBB) tissues.

[0078] In one embodiment, a biological tracer is used as an investigative and/or diagnostic tool to mark cells and/or to generate a specific animal model in vivo, and/or the biological tracer is applied for diagnostic instruments.

[0079] In one embodiment, the one or more inserted foreign DNA sequences are genes encoding proteins, peptides and transcription factors including IGF-1 (insulin

growth factor-1), insulin, rat growth hormone, glucocerebrosidase, brain-derived neurotrophic factor (BDNF), β -endorphin, HSV-1 (herpes simplex virus), anti-glioma, ApoE4, presenilin 1 and/or 2, myelin oligodendrocyte glycoprotein (MOG), crry, gata4, glutamate decarboxylase 65, interleukin-10, chondroitinase ABC, P/NK-1R, human interleukin-12 (hIL-12), stem cell transcription factors for iPSC including October 3/4, Sox2, Klf1-5, Nanog, LIN28 and Glis1, direct dopaminergic neurons reprogramming including ASC1, LMX1A, MYT1L and BRN2, direct glutamatergic, GABAergic neuronal reprogramming including ASC1, MYT1L, BRN2. and NeuroD1, direct retinal pigment epithelium reprogramming including PAX6, RAX, CRX, MITF-A, OTX2, NRL and KLF4/2, and direct spinal motor neurons reprogramming including NGN2, SOX11, ISL1 and LHX3, and further including anti-Tau, anti- β -amyloid antibodies, and nanobodies (VHH).

[0080] In one embodiment, the one or more inserted foreign DNA sequences include CRISPR/Cas9, antisense, sense, or mRNA used for treating a central nervous system (CNS) disease.

[0081] In one embodiment, the one or more inserted foreign DNA sequences include antisense or sense used for treating a central nervous system (CNS) disease, the CNS disease including brain viral infection.

[0082] In one embodiment, the one or more inserted foreign DNA sequences include shRNA, RNAi, or mRNA used for treating a central nervous system (CNS) disease, the CNS disease including brain viral infection.

[0083] In one embodiment, the expression vector is used to treat a central nervous system (CNS) disease including brain viral infection by delivering the hGH and the plurality of therapeutic genes via crossing the BBB.

[0084] In one embodiment, delivering the hGH and the plurality of therapeutic genes into the CNS via crossing the BBB is administrated with intravenous injection formulation, nasal spray formulation, inhalation formulation, liquid and/or tablet formulations, eye drop administration formulation, or a combination thereof.

[0085] In one embodiment, the regulator element includes (1) a tetracycline response element (TRE) including sequences of "TCCCTATCAGTGATAGAGA" to be included into the vector; and/or (2) a post-transcriptional regulatory element (PRE)+derived from woodchuck hepatitis virus (WHV) into 3' untranslated region of the AAV vector or lentiviral vector to enhance both titer and transgene expression.

[0086] In one embodiment, the intracellular or extracellular secretion sequences include fusion sequences, the fusion sequences including ROIKIWFONRR, YGRKKRRORRR, and/or LLNFDLLKILLAGDWESNPCP.

[0087] According to various embodiments of the present disclosure, a method for generating an expression vector for amplified expression in mammalian cells is provided. The method includes fusing one or more inserted foreign DNA sequences with intracellular or extracellular secretion sequences, the one or more inserted foreign DNA sequences including human growth hormone (hGH) and a plurality of therapeutic genes; configuring a promoter to drive expression of the one or more inserted foreign DNA sequences; configuring an insulator for stable and safety expression of the one or more inserted foreign DNA sequences; configuring a marker gene to encoding one or more corresponding proteins to indicate the expression of the one or more inserted foreign DNA sequences; and configuring a regulator element control expression of the hGH and the plurality of therapeutic genes under a controllable condition.

[0088] Various embodiments of the present disclosure provide detailed description of adeno-associated vector construction, cell culture and transformation, AAV production and purification, lysing of 293 cells and harvesting of rAAVs, purification of AAVs by ultracentrifugation, dialysis for in vivo application, heparin column purification of rAAVs, animal study, delivery of hGH gene into rat CNS by using the AAVs, brain immunofluorescence assay, immunocytochemical detection of cells, RNA extraction, construction of cDNA libraries, gene expression profiling with real-time PCR, protein sample preparation, hGH ELISA assay, and statistical analysis hereinafter.

[0089] In an exemplary experiment, for constructing adeno-associated vector encoding hGH and GFP, the cDNA of human growth hormone (hGH) with 598 bp was cloned into an AAV-2 vector (Applied Biological Materials Inc/abm) following enzymatic digestion by BamH-I and Hind III at 37° C. for 2 hours (New England Biolab). The

sequence of GFP was isolated from pUC12 vector (Clontech) by digestion of Hind III and Cla I, and then subcloned into the AAV-2 vector (abm). The backbone of this new construct may contain a CMV promoter, a polyadenylation site, and an ampicillin resistance gene.

[0090] In an exemplary experiment, for cell culture and transformation, human brain choroid plexus, HeLa and COS-7 cell lines were obtained from the American Type Culture Collection. COS-7 and HeLa cells were grown in DMEM (GIBCO) containing 10% heat-inactivated FBS. Human endothelial cells were grown in F12 K-medium with 2 mM L-glutamine containing 1.5 g/L sodium bicarbonate, 100 mg/mL heparin, 30 mg/mL endothelial cell growth supplement (ICN), and 10% FBS. The brain choroid plexus cells were grown in

[0091] Eagle's minimum essential medium with 0.1 mM nonessential amino acids, 90% Earle's balanced salt solution, and 10% FBS. H9 cells were grown in 80% RPMI 1640 medium (GIBCO) containing 10% FBS, 1 mM L-glutamine, 0.05 mg/mL gentamicin and 10,000 units/mL Penstrep (GIBCO). Cells were infected in DMEM/FBS containing 4 mg/mL Polybrene for 4-16 h. Human Embryonic Kidney HEK293 cells were cultured and maintained in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum at 37° C. in a fully humidified atmosphere of 5% CO₂. The cell culture condition may be modified based on actual needs, which may not be limited according to various embodiments of the present disclosure.

[0092] In an exemplary experiment, lipofectamine (Invitrogen) transfection kit was used for transfection Protocol. Other transfection kits or manners, commercial or noncommercial, may also be used based on actual needs, which may not be limited according to various embodiments of the present disclosure.

[0093] In an exemplary experiment, for adherent cells, 0.5-2×10⁵ cells were plated in 500 μL of growth medium without antibiotics one day before transfection, such that cells may be 90% confluent at the time of transfection.

[0094] In an exemplary experiment, for suspension cells, 4-8 ×10⁵ cells were plated in 500 μL of growth medium without antibiotics prior to preparing complexes.

[0095] In an exemplary experiment, for each transfection sample, complexes were prepared as the following. At step A, DNA was diluted in 50 μL of Opti-MEM® I Reduced Serum Medium without serum (or other medium without serum) and the mixture was mixed gently. At step B, Lipofectamine™ 2000 was gently mixed before use, then an appropriate amount was diluted in 50 μL of Opti-MEM® I Medium, which was incubated for 5 minutes at room temperature. Step C was proceeded within 25 minutes. At step C, after the 5-minute incubation, the diluted DNA was combined with diluted Lipofectamine™ 2000 (total volume is 100 μL). The mixture was mix gently and incubated for 20 minutes at room temperature (solution may appear cloudy). Complexes were stable for 6 hours at room temperature.

[0096] 100 μL of complexes was added to each well containing cells and medium and mixed gently by rocking the plate back and forth. Cells were incubated at 37° C. in a CO₂ incubator for 18-48 hours prior to transgene expression test. Medium may be changed after 4-6 hours.

[0097] In an exemplary experiment, for AAV production and purification, in the preparation of human embryonic kidney 293 cell and transfection by the AAV vector, 2×10⁶ cells were plated in ten culture plates (15 cm), and cells may

be 70-80% confluent before transfection (approximately 24 hours after plating). The cells were cultured in 20 mL of standard Dulbecco's modified Eagle medium (DMEM) with low glucose containing (10%) fetal calf serum and 100 µg/mL penicillin/100 µg/mL streptomycin. 3 hours before transfection, DMEM was removed and replaced with medium containing 5% fetal calf serum. The transfection reaction in each plate may contain 12 µg of adenovirus helper plasmid, 6 µg of the cap and rep plasmid (for example, AAVRH10), and 6 µg of the vector containing transgene of hGH (as shown in FIG. 1) by the protocol of calcium phosphate (2M CaCl_2 and Hepes buffer). After 24 hours transfection, 20 mL of fresh medium with 10% fetal calf serum were added into the dishes until viral harvests.

[0098] In an exemplary experiment, for lysing of 293 cells and harvesting of rAAVs, 48-72 hours after transfection, media was removed from cell culture plates and discarded. The cells were gently washed in warm 1× phosphate buffered saline (PBS at pH 7.4). 15 mL warm PBS was added to each plate, and the cells were gently removed with a cell scraper. The suspension was collected in 50 mL tubes. The cells were pelleted at 800×g for 10 minutes. The supernatant was discarded, and the pellet was resuspended in 11 mL of TD buffer (150 mM NaCl, 5.0 mM KCl, 0.7 mM K_2HPO_4 , 25.0 mM Tris pH 8.0) in a 50 mL tube. The 50 mL tube containing the resuspend solution was placed into the dry ice for 10 min, thawed for 15 min in 37° C. water bath, and vortexed for 2 min for repeating this procedure twice. A fresh solution of 10% sodium deoxycholate in dH₂O was prepared, and 1.25 mL of prepared fresh solution was added to each tube for a final concentration of 0.5%. Benzonase nuclease was added to a final concentration of 50 units per mL. Cellular debris was removed by centrifuging at 3000×g (1000 rpm) for 15 mins and transferred to fresh 50 mL tube. At this point, the sample may be stored at -20° C. before continuing its purification.

[0099] In an exemplary experiment, for purification of AAVs by ultracentrifugation, a concentration of 0.55 g CsCl/mL was added into the lysate solution in the 50 mL tube; after incubation for 10 min at room temperature, 5 µL of the final solution was taken for measuring its refractive index of 1.372 by a refractometer (AR200, Reichert Inc). 10 mL of final solution was loaded on a SW41 rotor that had spun for 48 hours at 38000 rpm at room temperature. The gradient solution was collected in 0.5 mL aliquots, and its refractive index at the range of 1.371-1.372 that are expected the highest titers (i.e., AAV2 and AAV5) in each fraction was determined and collected.

[0100] In an exemplary experiment, for dialysis for in vivo application, the fractions collected from CsCl gradient centrifugation were transferred in a dialysis cassette (Slide-A-Lyzer 10K, Thermo Scientific Inc) using a 21G 1½ plastic hub syringe. The cassette containing the solution was placed in a volume of 500 mL of 0.9% NaCl under the constant stirring for 2×60 min or overnight without stirring at room temperature. After the dialysis, samples were recovered by a 21G 1½ plastic hub syringes and were then aliquoted and stored at -80° C. until use or for titer determination by Q-PCR.

[0101] In an exemplary experiment, for heparin column purification of rAAVs, the Hi Trap heparin column was set up using a peristaltic pump, so that solutions flowed through the column at 1 mL per minute without introducing air bubbles into the heparin column. The column was equili-

brated with 10 mL 150 mM NaCl and 20 mM Tris at pH 8.0. The 50 mL virus solution was applied to the column by flowing through the column. The column was washed with 20 mL 100 mM NaCl and 20 mM Tris at pH 8.0. A 5 mL syringe was used to continue to wash the column with 1 mL 200 mM NaCl and 20 mM Tris at pH 8.0, followed by 1 mL 300 mM NaCl and 20 mM Tris at pH 8.0; and the flow-through was discarded. The 5 mL syringe and gentle pressure were used to elute the virus from the column by applying: 1.5 mL 400 mM NaCl and 20 mM Tris at pH 8.0; 3.0 mL 450 mM NaCl and 20 mM Tris at pH 8.0; 1.5 mL 500 mM NaCl and 20 mM Tris at pH 8.0; and the column eluate were collected in a 15 mL centrifuge tube.

[0102] In an exemplary experiment, the vector was concentrated using Amicon ultra-4 centrifugal filter units with a 100,000 molecular weight cutoff. 4 mL of column eluate was loaded into the concentrator and centrifuged at 2000×g for 2 minutes (at room temperature). Flow through was discarded, and the concentrator was reloaded with remaining virus solution with repeat centrifugation. The concentrated volume may be approximately 250 µL. If the concentrated volume is significantly more than 250 µL, the flow through was discarded, and the centrifugation was continued in one minute step until the volume is approximately 250 µL. 250 µL of PBS was added to virus for a final volume of 500 µL and removed from the concentrator. The vector was filtered through a 13 mm diameter (0.2 µm) syringe filter. The vector may be aliquoted and stored at -80° C. as needed.

[0103] In an exemplary experiment, adult rats (250 g), obtained from Taconic Farm (Ejby, Denmark), were maintained in an animal facility of Uppsala University in a temperature and light-controlled room, with food and water available ad libitum. Before the AAVs construct injection, the rats were anesthetized prior to viral vector injections. Rats were placed in a small-animal stereotactic apparatus fitted with a rat adaptor that positioned the skull horizontally between lambda and bregma. Animal experiments were approved by the Animal Care and Use Committee of Uppsala University at Biomedical Center.

[0104] In an exemplary experiment, for delivery of hGH gene into rat's CNS by using the AAVs, rats may be divided into four groups (6 animals per group); the first and second groups may receive AAVRH10 vector only via either the CNS direct injection or tail-vein injection, respectively; and the third and fourth groups may receive a combination dosage of AAVRH10-GFP at ⅓ of the total volume, and AAVRH10-hGH at ⅔ of the total volume via either the CNS direct injection or tail-vein injection, respectively. The vectors may have the conditions as following: one vector may contain both genes of hGH and GFP in one delivery system; and two individual vectors may be the 1st vector containing hGH only and the 2nd vector containing GFP only which may be delivered at the same time. The animals were anesthetized and the head was fastened in a stereotactic apparatus. Injections of viral vectors into the mouse brain were performed at the following coordinates: for injection into the hippocampus, 4.7 mm interaural to bregma, 3.5 mm to the right of the midline, and 3.0 mm depth. Three microliters of concentrated viral vectors were loaded into an internal cannula needle (C315× 33) with cannula tubing connected to a Hamilton syringe mounted onto a microinjection pump (Harvard Apparatus, Dover, MA). The viral vector solutions were delivered at a rate of 0.5 µL/min. For tail-vein injection, 100 µL of viral vectors was used.

[0105] In an exemplary experiment, for brain Immuno-fluorescence assay, animals were killed 2 months after injections, and the brains were dissected as the following. The whole brain was divided as two parts, one part, as intact tissue of half brain, was frozen in the dry ice immediately, another half was dissected into different parts of brain, including frontal cortex, cortex, hippocampus, striatum, and liver. Each of these samples was further divided into two pieces for both RNA extraction and protein preparation, respectively. The brain tissues were embedded in O.C.T. (optimum cutting temperature) medium (Tissue-Tek, Miles Inc., Indianapolis, USA) and frozen in a methanol/dry ice bath. The frozen tissues were sectioned at a thickness of 15 μ m per coronal section using a cryostat (Bright Instrument, Huntingdon, U.K.) at -18° C.

[0106] In an exemplary experiment, for immunocyto-chemical detection of cells (e.g., neurons in CNS), the slides were fixed with 4% paraformaldehyde containing 4% sucrose in PBS for 10 min, and washed three times with the PBT buffer (PBS in 1xHSS, 0.1% bovine serum albumen [BSA] and 0.2% Tween 20), then blocked with 10% goat serum for 15 min. After washing three times with the PBT buffer, slides were incubated with the primary specific antibody of human growth hormone (1:500, R&D System Inc) at 4° C. overnight. The secondary antibodies conjugated with tetramethylrhodamine isothiocyanate (TRITC) with red fluorescent light (1:800) under the mounting medium of Northern Light (R&D System Inc) were then added onto slides for 30 min at room temperature. The slides were washed with the PBT buffer and analyzed by using a Zeiss 510 (Zeiss, Inc. Thornwood, NY) fluorescent microscope.

[0107] In an exemplary experiment, for RNA extraction, samples used for the gene expression or microarray study were immediately placed in RNeasy Lysis Buffer (Qiagen) for RNA extraction. Tissue samples were homogenized with a Homogenizer (OMNI-TH International Inc). The samples were mixed using the autoclaved 0.5 mm stainless tip in a tube containing 600 μ L of the QIAzol lysis reagent (Qiagen), homogenized for 2 min and then placed on ice. The total RNA was extracted with a RNeasy Mini Kit (Qiagen) according to the manufacturer's recommendations. The quality of RNA was measured using a 2100 Bioanalyzer (Agilent Technologies, Palo Alto, CA). The samples which reached a 28S/18S ribosomal RNA ratio of 1.7 with RIN score >6.5 were kept for further steps on the gene expression by Q-PCR and/or microarray. Briefly, 550 μ L of RNA Nano gel was loaded into a spin filter and centrifuged at 1500 rpm for 10 min at room temperature; 65 μ L of the gel was mixed with 1 μ L of Nano 6000 dye and centrifuged at 13000 g for 10 min at RT; 9.0 μ L of this gel-dye mixture was then loaded into 3 wells marked with G of the RNA Nano Chips (Agilent) and 5.0 μ L of RNA 6000 Nano Market was loaded into all 12 sample wells, subsequently; and 1.0 μ L of samples were added into each sample well. The chip was then placed horizontally in the adapter of the IKA vortexes and vortexed for 1 min before loading into the Agilent 2100 bioanalyzer.

[0108] In an exemplary experiment, for construction of cDNA libraries, the total rat RNA (500 ng) was reverse transcribed using a SuperScript (VLO™) First-Strand cDNA synthesis kit according to the manufacture instruction (Invitrogen). The reaction component included 10x SuperScript Enzyme mix and 5xVLO Reaction Mix including random primers, $MgCl_2$, and dNTPs, and the tubes were

subjected into a PCR program of 25° C. for 10 min, 42° C. for 60 min, and 85° C. for 5 min. The final samples of 1st strand cDNA were stored at -20° C. until real-time PCR is needed. Expression was further validated by QPCR using a (2x) Taqman Universal PCR Master Mix (Applied Biosystem Inc). The cDNA was diluted to have the final concentration of 1.0 ng/ μ L.

[0109] In an exemplary experiment, for gene expression profiling with real-time PCR, the 1st strand of cDNA synthesized from total RNA was used as template for real-time PCR. The reaction was carried out on an optional tube including 10 ng of the synthesized cDNA, 10 μ L of a (2x) Universal PCR Master Mix (Applied Biosystem Inc) and 1.0 μ L of TaqMan Probes purchased from Applied Biosystem Inc with a total final volume of 20 μ L. The real-time PCR reactions were run on the Instrument (Bio-Rad).

[0110] In an exemplary experiment, for protein sample preparation, after the tissues from rat brain and liver were dissected, dissected tissues were frozen immediately on dry ice and stored at -80° C. For preparing enzyme-linked immunoassay (ELISA) samples, the individual sample (100 μ g) was collected and crushed on dry ice and placed into a 5.0 mL of pre-chilled tube with 1.0 mL of cold PBS buffer containing 10 μ L of protease inhibitors (Sigma). The tissue then was homogenized at a low speed for about 20 seconds per time for 3 times and kept cool on wet ice. The suspended solution was then transferred into a 1.7 mL of microcentrifuge tube and spined at 14,000 rpm at 4° C. for 15 min. The supernatant was aliquoted and stored at -80° C. 50 μ L of the solution was diluted with 1:10 before running the ELISA.

[0111] In an exemplary experiment, for the hGH ELISA assay, the samples were diluted to equal concentration and then the ELISA was run using the human growth hormone ELISA kit (R & D System Inc). Brief description, 50 μ L of standard, control or sample was added in each well (96 wells). The well was covered with the adhesive strip provided and incubated for 2 hours at room temperature. Each well was aspirated and washed with 400 μ L of wash-buffer for three times for a total of four washes. Complete removal of liquid at each step is essential to desirable performance. After the last wash, any remaining wash-buffer was removed by aspirating or decanting, and the plate was inverted and blotted against clean paper towels. 200 μ L of GH Conjugate was added to each well, which was covered with a new adhesive strip and incubated for 2 hours at room temperature. After washing step for three times, another 200 μ L of substrate solution was added to each well and incubated for 30 minutes at room temperature under the dark. At the end, 50 μ L of stop solution was added into each well. The color in the wells should change from blue to yellow. The plate was ready for determining the optical density of each well within 30 minutes using a Microplate Reader (Molecular Devices, LLC) set to 450 nm.

[0112] In an exemplary experiment, statistical analyses were performed using the software Prism (Graphpad Software Inc) and Excel program (Microsoft Inc). For the ELISA assay, the computer program of SortMax Pro Software (Molecular Devices, LLC) was used to test for a significant level of enhancement of antibody responses; the Kruskal-Wallis One-Way ANOVA was performed; and P-values less than 0.05 were considered significant. For gene expression, data was analyzed by using the computer program of GeneSpring (GX 11) including pre-processing raw-data, normalize data, QC samples/entities, t-test plus Ben-

jamin correction, clustering, annotation and access biological context. Comparisons of biological functional pathway or biomarkers were run by an IPA program (Ingenuity® systems) and Metacore (GeneGo) program.

[0113] The vector may be constructed in various embodiments of the present disclosure. In order to check the ability of the constructed AAV vector to transduce brain cells, the cells derived from the choroid plexus may be used. The AAV-2 vector containing the hGH gene may be tested in vitro together with the AAV-2 vector expressing GFP. Results may demonstrate a clear and distinct expression of GFP in these cells (FIGS. 2A-2B). In the following in vivo experiment, one group of rats may be given the AAVrh10-hGH vector together with the AAVrh10-GFP vector via a tail-vein injection and the other group may receive the AAVrh10-hGH and AAVrh10-GFP vectors via an intra-hippocampal injection.

[0114] FIG. 3A depicts an exemplary fluorescent image of intra-hippocampal injection of recombinant AAV vectors into the hippocampus of a rat brain according to various disclosed embodiments of the present disclosure; FIG. 3B depicts an exemplary fluorescent image of intra-hippocampal injection of recombinant AAV vectors without any transgenes (as the control) into the hippocampus of a rat brain according to various disclosed embodiments of the present disclosure; and FIG. 3C depicts an enlarge view of a boxed region in FIG. 3A. FIG. 3A may show the expression of transgenes (fluorescent light indicated by the white arrows) in hippocampus of rat brain after in vivo intra-hippocampal injection of 5.0 μ L of recombinant AAV vectors with the combination dosage of $\frac{2}{3}$ of rAAV-hGH (3.5 μ L) and $\frac{1}{3}$ of rAAV-GFP (1.5 μ L). Referring to FIG. 3B, the control group was injected with 5.0 μ L AAV vector only. FIG. 3C illustrates a large view of injection site indicating significant GFP expression of transgenes in the region of hippocampus. Referring to FIG. 3C, the boxed region may show various CNS cells transduce by the transgenes from recombinant AAVs delivery. The slides were analyzed using a fluorescent microscope (10 $\times$).

[0115] FIG. 4A depicts an exemplary fluorescent image from the expression of transgenes (fluorescent light indicated by the white arrows) in CNS tissues after tail-vein injection of recombinant AAV vectors encoding transgenes of GFP and/or human growth hormone hGH in animal models according to various disclosed embodiments of the present disclosure; and FIG. 4B depicts an exemplary fluorescent image in a control animal model after tail-vein injection of recombinant AAV vectors without any transgenes according to various disclosed embodiments of the present disclosure.

[0116] FIG. 4A shows the expression of transgenes in hippocampus of rat brain after the tail-vein injection of 100 μ L of recombinant AAV vectors with the combination dosage of $\frac{2}{3}$ of rAAV-hGH (67 μ L) and $\frac{1}{3}$ of rAAV-GFP (33 μ L). For FIG. 4B, the control group was injected with 100 μ L AAV vectors only. The slides (20 $\times$) were analyzed using a fluorescent microscope.

[0117] FIG. 5A depicts an exemplary quantitative-polymerase chain reaction (qPCR) detection result of mRNA expressional levels of human growth hormone hGH after the tail-vein injection according to various disclosed embodiments of the present disclosure; and FIG. 5B depicts an exemplary quantitative-PCR (qPCR) detection result of mRNA expressional levels of human growth hormone hGH

after intra-hippocampal injection according to various disclosed embodiments of the present disclosure. The total RNAs were extracted from different parts of rat brain and liver, and individual cDNA library was generated by reverse transcribed (RT) reaction using a SuperScript (VLO™). Subsequently, the qPCR reaction was performed using a specific Taqman probe of human growth hormone (ABI).

[0118] In various embodiments of the present disclosure, the rat mRNA expression of hGH after intra-hippocampal and tail vein injections may be evaluated. The mRNA expression of hGH may be analyzed in four different areas of the brain (hippocampus, frontal cortex, cortex and striatum) and in the liver (FIG. 5A). Compared to the control rats, the results may demonstrate an increased mRNA expression of hGH in the liver of the tail-vein injection group. An increased hGH mRNA expression in hippocampus, frontal cortex, cortex, and striatum may also be detected in the tail-vein injection group (FIG. 5B). In the CNS injection group, compared to the control rats, an up-regulated mRNA expression of hGH may also be observed in all four brain regions. In addition, hGH mRNA expression may also be observed in the livers of these rats; and no expression of hGH may be detected in control animals.

[0119] FIG. 6A depicts an exemplary quantitative-PCR (qPCR) detection result of mRNA expressional levels of rat IGF-1 after the tail-vein injection according to various disclosed embodiments of the present disclosure; and FIG. 6B depicts an exemplary quantitative-PCR (qPCR) detection result of mRNA expressional levels of rat IGF-1 after intra-hippocampal injection according to various disclosed embodiments of the present disclosure. The cDNA library was generated from the total RNA extracted from brain tissue and liver by reverse transcribed (RT) reaction using a SuperScript (VLO™). Subsequently, the qPCR reactions were performed using a Taqman probe of rat IGF-1 (ABI). FIG. 6A may show an up-regulation after the tail-vein injection compared to the control group. FIG. 6B may demonstrate that the mRNA level of hGH in the rats given an intra-hippocampal injection was significantly up-regulated compared to the control group.

[0120] For the mRNA expression of rIGF-1 in various embodiments of the present disclosure, the mRNA expression of rIGF-1 may be analyzed in hippocampus, frontal cortex, cortex, striatum and in the liver. Compared to the controls, the results may demonstrate an increased mRNA expression of IGF-1 in hippocampus, frontal cortex, cortex striatum, and liver in the tail-vein injected rats (FIG. 6A). In the CNS injection group, compared to the controls, an up-regulated expression of IGF-1 mRNA may be found in all four brain regions analyzed (FIG. 6B). A small increase of IGF-1 mRNA expression compared to the controls may also be found in the liver.

[0121] FIG. 7A depicts an exemplary fluorescent image of neurons detected by containing a specific anti-NeuN antibody with IgG conjugated to TRITC (A594/red fluorescent, 20 $\times$), where the slide was sectioned from CNS samples after the tail-vein injection, according to various disclosed embodiments of the present disclosure; and FIG. 7B depicts an exemplary fluorescent image for co-expression of hGH in the neurons by using a specific anti-human growth hormone antibody with IgG conjugated to FITC (A488/green fluorescent light, 20 $\times$), where the slide was sectioned from CNS samples after the tail-vein injection according to various disclosed embodiments of the present disclosure.

[0122] For FIG. 7A, the neurons were illustrated by a cocktail solution containing a specific anti-NeuN antibody with IgG conjugated to TRITC (A594/red fluorescent, 20×). For FIG. 7B, the expression of hGH was co-localized in the same area as the neurons, shown in FIG. 7A using a specific anti-hGH antibody with IgG conjugated to FITC (A488/green fluorescent, 20×). The slide was sectioned from samples of the tail-vein injection, stained by a cocktail solution containing double antibodies, covered by DAPI medium, and analyzed using a confocal microscope (Olympus Fluoview 1000 system).

[0123] In various embodiments of the present disclosure, co-localization of NeuN and hGH in the CNS tissues may be evaluated. Brain sections from the rats given a tail-vein injection with AAVrh10-hGH may be analyzed using immunohistochemistry. The sections may be labelled with antibodies against both neuronal marker protein of NeuN (FIG. 7A) and the hGH protein (FIG. 7B) in the hippocampus region.

[0124] The results may demonstrate co-localization of neurons labelled with NeuN and the hGH protein (FIGS. 8A-8C). FIG. 8A depicts an exemplary fluorescent image of neurons for a specific neural biomarker protein which was expressed in neurons of the CNS and detected by anti-NeuN antibody (white arrows indicate red fluorescent light), where the slide was sectioned from the CNS samples after the tail-vein injection, according to various disclosed embodiments of the present disclosure; FIG. 8B depicts an exemplary fluorescent image for the expression of hGH proteins in the cytoplasm of the neurons, detected by a specific anti-human growth hormone antibody (white arrows indicate green fluorescent staining/light), where the slide was sectioned from the CNS samples after the tail-vein injection, according to various disclosed embodiments of the present disclosure; and FIG. 8C depicts an exemplary overlapped view of FIG. 8A and FIG. 8B.

[0125] FIGS. 8A-8C illustrate co-localization of neurons with expression of hGH by double antibody-labelled immunostaining. For FIG. 8A, the neurons were illustrated by a cocktail solution containing two antibodies, one of them was a specific anti-NeuN antibody with IgG conjugated to TRITC (A594/red fluorescent, 40×). For FIG. 8B, the expression of hGH was co-localized in the same area as the neurons using a specific anti-hGH antibody with IgG conjugated to FITC (A488/green fluorescent, 40×). The data may indicate that the neural marker protein was expressed in neurons of the CNS (white arrows in FIG. 8A), and the expression of hGH proteins was only in the cytoplasm of the neurons (white arrows in FIG. 8B). The brain was sectioned from rats given the tail-vein injection, stained by a cocktail

solution containing double antibodies, covered by DAPI medium, and analyzed using a confocal microscope (Olympus Fluoview 1000 system).

[0126] FIG. 9A depicts an exemplary fluorescent image for the expression of hGH proteins detected by a specific antibody (green fluorescent staining/light) in CNS choroid plexus, neurons identified by a specific anti-NeuN antibody (red fluorescent light), and overlapped co-localizations (yellow fluorescent light) of both hGH proteins and neurons in a same view of the tissues with the delivery of recombinant AAV vectors after tail-vein injection according to various disclosed embodiments of the present disclosure; FIG. 9B depicts an exemplary fluorescent image for a control group without hGH protein expression according to various disclosed embodiments of the present disclosure; and FIG. 9C depicts an exemplary fluorescent image for the expression of hGH proteins in another section of CNS choroid plexus tissue with the delivery of recombinant AAV vectors after tail-vein injection according to various disclosed embodiments of the present disclosure.

[0127] For FIG. 9A, the hGH proteins were detected by a cocktail solution containing two antibodies, including a specific anti-hGH antibody with IgG conjugated to FITC (A488/green fluorescent, 10×). The high level of the hGH proteins was significantly observed on the choroid plexus tissues, and overlapped with the neurons; these neurons were identified by a specific anti-NeuN antibody with IgG conjugated to TRITC (A594/red fluorescent, 10×) in both sections of A and B. FIG. 9B is served as a control (group) in same tissues of the CNS brain without expression of hGH protein. The brain was sectioned from rats given the tail-vein injection, stained by a cocktail solution containing double antibodies, covered by DAPI medium, and analyzed using a confocal microscope (Olympus Fluoview 1000 system).

[0128] It should be noted that high levels of the hGH expression (FIG. 9A, green fluorescent) may also be observed in the choroid plexus. In contrast, no detectable levels of these constituents may be found in corresponding regions of control animals (FIG. 9B).

[0129] In various embodiments of the present disclosure, the protein levels of hGH in hippocampus, frontal cortex, cortex, striatum, and in the liver may be analyzed with ELISA. In the rats given the intra-hippocampal injection with AAVRH10-hGH, expression of hGH may be observed in all four brain regions analyzed, with the highest expression in hippocampus (as shown in Table 3). Table 3 shows the Immunoassay human growth hormone levels. hGH expression may also be detected in the liver of these rats but to a lower extent. In the rats given the tail-vein injection with the AAVRH10-hGH vector, a high expression of hGH may be observed in the liver tissue. A significant expression of hGH may also be detected in the hippocampus, frontal cortex, cortex, and striatum (as shown in Table 3).

TABLE 3

	Tail-vein		Intracerebral	
	Controls (pg/ml)	hGH (pg/ml)	Controls (pg/ml)	hGH (pg/ml)
Hippocampus	0.13 ± 0.03	165.62 ± 14.57	0.16 ± 0.03	1669.74 ± 43.11
Frontal cortex	0.15 ± 0.03	118.57 ± 33.37	0.16 ± 0.03	649.39 ± 33.53
Cortex	0.09 ± 0.01	194.52 ± 7.88	0.13 ± 0.02	861.56 ± 145.80
Striatum	0.07 ± 0.01	138.28 ± 26.93	0.08 ± 0.02	585.50 ± 17.66
Liver	0.25 ± 0.06	1067.84 ± 75.40	0.24 ± 0.04	8.12 ± 1.17

[0130] According to various embodiments of the present disclosure, the delivery of hGH into CNS may be able to act its biological function in both areas of CNS and peripheral system. The samples collected from brain tissues and liver may be prepared as a protein solution in a 96 well plate. Determination of hGH level may be performed according to the protocol of hGH ELISA kit (R&D System Inc) and the optical density of each well may be measured using a Fluorostar microplate reader (Molecular Devices, LLC) set at 450 nm. As shown in the Table 3 from top to the bottom are hippocampus, frontal cortex, cortex, striatum, and liver. X-axis may indicate diluted tissue samples [AG65] (1:50); and Y-axis may show the concentration of hGH level (pg/mL).

[0131] Body weight may be described in various embodiments of the present disclosure herein. To confirm the effect of the transferred hGH gene on its protein product, the body weight of the rats may be monitored throughout the experiment. FIG. 10A depicts effect of hGH treatment in tail-vein injection on body weights according to various disclosed embodiments of the present disclosure; and FIG. 10B depicts effect of hGH treatment in CNS injection on body weights according to various disclosed embodiments of the present disclosure. The weight gain may be calculated by subtracting the recorded weight at the first day from the weight at the final day of the experiment. Both the tail vein and the CNS injection groups receiving the AAVRH10-hGH vector may gain significantly more weight than the rats injected with the control vector (FIGS. 10A-10B). The change in body weight may be significantly higher ($p < 0.01$) in the group given the tail-vein injection with AAV-hGH compared to its corresponding control group (FIG. 10A). In addition, animals treated with AAV-hGH by the intra-hippocampal injection may display a significantly increase ($p < 0.05$) in body weight gain compared to its corresponding control group (FIG. 10B). The increase in body weight may depend on the route of vector administration as the recorded weight gain following the tail-vein injection may slightly exceed that obtained following direct intra-hippocampal injection.

[0132] The objective of the present disclosure may be to determine a desirable route for the hGH administration targeting the brain by applying gene therapy. Various embodiments of the present disclosure may indicate that the vector carrying the hGH gene that is capable to cross the BBB may be successfully constructed. Thus, the vector may be suitable for reaching hGH-responsive areas not only through direct intra-cranial injection but also through the tail-vein injection. Furthermore, it may be observed that the transferred gene may be transcribed to its message, which was subsequently translated to the protein hormone in order to elicit its biological or therapeutic effects in the CNS. It may also be observed that the hGH mediator IGF-1 may be formed as a result of hGH stimulation.

[0133] Moreover, co-localization of neurons with the expression of hGH by double antibody-labelled immunostaining may be observed. Thus, the expression of hGH may be co-localized in the same area as the neurons according to the experiment where the specific anti-NeuN antibody with IgG conjugated to TRITC (A594/red fluorescent) may be co-localized in the same area as a specific anti-hGH antibody with IgG conjugated to FITC (A488/green fluorescent). The observation may be considered to be indicative of an expression of the hGH-gene in neurons of the CNS.

[0134] The choroid plexus (CP) along with the BBB is recognized for providing a robust protective effort for the CNS, which is a physical barrier to impede entrance of toxic metabolites to the brain. It has been attributed a function related to regulation of the transport of growth factors, hormones and other medicinal agents over the BBB, as well as its preponderant role on initiating central diseases and remediation. It should be noted that high levels of hGH proteins may be also found in CP (FIGS. 9A-9C). FIG. 9A may indicate the presence of hGH binding protein on the receptor, as detected by its specific antibody (green fluorescent) in animals receiving the AAVRH10-hGH vector, whereas no binding was found in the control animal (FIG. 9B). The observation may reflect that hGH formed in the CNS may find and bind to the sites in the CP. However, hGH binding to its receptor sites in the CP may also reflect an overproduction of the hormone in peripheral tissues, such as the liver.

[0135] A further observation confirming a successful transfer of the hGH gene by the presence of the hormone protein in the rat may be indicated by the increased body mass following the injection of the AAVRH10-hGH vector as shown in FIGS. 10A-10B. Compared to the control, both CNS and tail-vein injections may cause a significant increase in body weight gain. The body weight increase following the peripheral injection through the tail vein may surpass the body weight increase following the direct CNS administration into the hippocampus.

[0136] Regarding the detected levels of immuno-reactive hGH which are recorded by the ELISA technique, it should be noted that the hormone protein may be expressed in most brain regions examined in both intra-cerebral injection and tail vein injection. In both routes of administration of the AAVRH10-hGH vector, the hormone may be found to be detected in areas like hippocampus and liver. Compared to the controls, for the tail-vein injection, it may detect increased amounts of hGH in the brain areas hippocampus, frontal cortex, cortex and striatum.

[0137] Similar observation may be seen in the case of CNS injection, where the hormone levels may be significantly higher in all brain areas examined, particularly in the hippocampus. The detected hGH level in the liver of rats exposed to the peripheral administration of the AAVRH10-hGH vector may be significantly high compared to the expected very low levels in rats that received the intra-cerebral injection.

[0138] According to various embodiments of the present disclosure, it may demonstrate that current AAV vector may be used to transfer the hGH gene from the periphery into the brain area hippocampus. The hormone protein may be expressed and functionally active as concluded from the observation that the expression of the gene transcript of its mediator IGF-1 may be stimulated. However, it should be noted that although the hormone can reach desired areas in the brain, it may also be expressed in non-desirable areas (e.g., the liver). The expression of the GH in the liver area and in other peripheral tissue may be related to the body weight increase as observed in experimental animals. An increase may be most pronounced in animals receiving the tail-vein injection.

[0139] According to various embodiments of the present disclosure, the results may suggest that the used delivery system may not only be helpful as an investigative tool for understanding of the function of the GH-related target cells

in the CNS, but also may be a valuable therapeutic implication on treatment of GH-deficient patients. However, it should be noted that the vector construct may need to be modified in a manner that the delivery of the GH gene may be limited to specific targets (e.g., relevant neurons) for its desirable hormone effect.

[0140] Therefore, the construct used herein may be further developed as universal delivery system to carry other gene products to target the diseases in the CNS. The carriers may be viral gene vectors such as AAV or HIV, fusion proteins, activated domain of polypeptides or peptides. Since the GH receptors are present in the human choroid plexus, it may suggest that hGH might enter into the brain via binding to its receptors in the tissues. The binding domain of such sequence may be selected and fused with other therapeutic protein or inserted on the surface of the envelope or capsule of the vector particle. In addition, expression or production of fusion proteins with capability to cross the blood brain barrier may be designed using the vector system. Such approach may be useful for the treatment of patients with CNS disorders and other suitable diseases.

[0141] Although it is evident some viral gene vectors that delivered transgenes or therapeutic genes into the cells of the CNS have been developed, these vectors may not be able to cross the BBB. Overcoming the difficulty of delivering therapeutic agents to specific regions of the brain may present a major challenge to treatment of most brain disorders. Therefore, in the present disclosure, the vector (e.g., platform) may be developed for delivering recombinant hGH as an example of therapeutic protein or peptide to target certain tissues or cells in the CNS by crossing the BBB, which may elicit desired and sufficient therapeutic effects in the target tissues and cells of the CNS in animal

models in vivo. The vector may have therapeutically implication for treatment of those patients who are suffered from neurodegenerative disorders such as those related to GH deficiency. These disorders may be from childhood onset of GHD but also from acquired GHD over the life span. For example, complications may appear during aging, but also may result from various trauma or exposure to addicted drugs. In brain injuries or trauma to the spinal cord, a decline in GH function may be repaired or restored following GH therapy, and GH therapy may also be useful for the treatment of cognitive decline resulting from long term exposure to opioids.

[0142] As disclosed above, the present disclosure provides a route for delivery of hGH to its responsive area in the brain by applying gene therapy. The AAV vector may be constructed to carry the hGH gene across the BBB. The AAV vector construct may demonstrate desirable features in experimental animals for the delivery of the gene of hGH to desired areas in the brain by not only following the intracranial injection but also following the tail-vein injection. It may be confirmed that the transferred gene may be transcribed to its mRNA and subsequently translated to the protein hormone, which may be capable of inducing the formation of its mediator IGF-1.

[0143] Although some embodiments of the present disclosure have been described in detail through examples, those skilled in the art should understand that the above-mentioned embodiments are only for illustration and not for limiting the scope of the present disclosure. Those skilled in the art should understand that the above-mentioned embodiments may be modified without departing from the scope and spirit of the present disclosure. The scope of the present disclosure may be defined by the appended claims.

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<213> ORGANISM: Homo sapiens

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ggagagaata ctgcccagtg gagaagccag gagaacgagg aggacggtga ggaggaccct	180
gaccgctatg tctgtagtgg ggttccggg cggccgccag gcctggagga agagctgacc	240
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gtggtggtag ccaccatcaa gtctgtgcgc ttctacacag agaagaatgg acagctcatc	360
tacacgccat tcactgagga cacaccctcg gtgggccagc gcctcccaa ctccgtgctg	420
aacaccctca tcatgatcag cgtcactgtg gttatgacca tcttcttggg ggtgctctac	480
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gcccaggaga gaaatgagcc catattccct gccctgatat actcatctgc catggtgtgg	900
acggttggca tggcgaagct ggaccctcc tctcagggtg ccctccagct cccctacgac	960
ccggagatgg aagaagactc ctatgacagt tttggggagc cttcataccc cgaagtcttt	1020
gagctccct tgactggcta cccaggggag gagctggagg aagaggagga aaggggcgtg	1080
aagcttggcc tcggggactt catcttctac agtgtgctg tgggcaaggc ggctgccacg	1140
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ctgacccctc tgctgcttgc tgtgttcaag aaggcgctgc ccgccctccc catctccatc	1260
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<210> SEQ ID NO 14

<211> LENGTH: 1323

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 14

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tgttcaggca gaaacttgct tacattacca tctggactgc aagagaatat tatacattta	180

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aacctgtctt ataaccactt tactgatctg cataaccagt taaccaata taccaatctg	240
aggaccctgg acatttcaaa caacaggctt gaaagcctgc ctgctcactt acctcggctc	300
ctgtggaaca tgtctgtgc taacaacaac attaaacttc ttgacaaatc tgatactgct	360
tatcagtggg atcttaaata tctggatgtt tctaagaaca tgctggaaaa ggttgctctc	420
attaaaaata cactaagaag tctcgagggt ctcaacctca gtagtaacaa actttggaca	480
gttccaacca acatgccctc caaactacat atcgtggacc tgtctaataa ttctttgaca	540
caaattcttc caggtagatt aataaacctg acaaatctca cacatcttta cctgcacaa	600
aataagtcca cattcattcc agaccaatct ttggaccaac tctttcagtt gcaagagata	660
accttttaca ataacagggt gtcattgtgac cacaacaaa acattactta cttactgaag	720
tggtgatggg aaacaaaagc ccatgtgata gggactccat gttctacca aatatcatct	780
ttaaaggaaac ataacatgta tcccacacct tctggattta cctcaagctt attcactgta	840
agtgggatgc agacagtggg caccattaac tctctgagtg tggtaaacta acccaaagt	900
accaaatac ccaacaata tcgaacaaag gaaacaacgt ttggtgccac tctaagcaaa	960
gacaccacct ttactagcac tgataaggct tttgtgccct atccagaaga tacatccaca	1020
gagactatca attcacatga agcagcagct gcaactctaa ctattcatct ccaagatgga	1080
atggtcacia acacaagcct cactagctca acaaaatcat cccaacacc catgacccta	1140
agtatcacta gtggcatgcc aaataatttc tctgaaatgc ctcaacaaag cacaaccctt	1200
aacttatgga gggagagac aaccacaaat gtaaagactc cattaccttc tgtggcaaat	1260
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tga	1323

<210> SEQ ID NO 15
 <211> LENGTH: 180
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 15

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cttaagtttg ctgttgagat atcttgcaaa tacaatgct ttcttgggta tatcaagagg	120
cattctttat cacctcccta gagatcttca agtggtcaga tgctcagcag ttctgtaaa	180

<210> SEQ ID NO 16
 <211> LENGTH: 1327
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 16

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acaccgctgg tgcctcctc cgtgctgggc ctgtcctacc tccagggcg aggcgctggc	180
tctgcgtccg gaggcgctc gggcggcagc tccggtgggg ccgcgtctgg tgcggggccc	240
gggacccagc agggcagccc gggatggagc caggcgggag ccgacggagc cgcttacacc	300
ccgcgcctgg tgctgcgcg cttctccttc ccggggacca ccgggtccct ggcgccgcgc	360
gcgcgcctg ccgcggcccc ggaagctgc gcctacagca gtggcgccgg agcgcggggt	420

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gcgggacctg	cgggccgcga	gcagtactgg	cgcgcggct	tcgcgggctc	ctaactccagc	480
ccctaccccg	cttacatggc	cgacgtgggc	gcgtcctggg	cgcagccgc	cgccgcctcc	540
gcgggccct	tcgacagccc	ggctctgcac	agcctgcccg	gccgggccc	cccgccgcgc	600
cgacacccca	atctcgatat	gtttgacgac	ttctcagaag	gcagagagt	gttcaactgt	660
ggggctatgt	ccaccccgct	ctggaggcga	gatgggactg	gtcactatct	gtgcaacgcc	720
tcgggcctct	accacaagat	gaacggcatc	aaccggccgc	tcacaaagcc	tcagcgccgg	780
ctgtccgcct	cccgcgagt	gggcctctcc	tgtgccaa	gccagaccac	caccaccacg	840
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ctccacgggg	tcccagggc	tcttgcaatg	cggaaagagg	ggatccaaac	cagaaaacgg	960
aagcccaaga	acctgaataa	atctaagaca	ccagcagctc	cttcaggcag	tgagagcctt	1020
cctcccgcca	gcggtgcttc	cagcaactcc	agcaacgcca	ccaccagcag	cagcgaggag	1080
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tcggccctga	agctctcccc	acaagctat	gcgtctcccg	tcagccagtc	tccacagacc	1260
agctccaagc	aggactcttg	gaacagcctg	gtcttggccg	acagtcacgg	ggacataatc	1320
actgcgt						1327

<210> SEQ ID NO 17

<211> LENGTH: 1758

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 17

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ggaaacaaa	tgtgcgcct	gctctacgga	gacgcgcgga	agccggcgga	gagcggcggg	180
agccaacccc	cgccggccgc	cgcccggaag	gccgcctcg	cctgcgacca	gaagccctgc	240
agctgctcca	aagtggatgt	caactacgcg	tttctccatg	caacagacct	gctgccggcg	300
tgtgatggag	aaaggccac	tttggcggtt	ctgcaagatg	ttatgaacat	ttactttcag	360
tatgtggtga	aaagtttcga	tagatcaacc	aaagtgattg	atttcatta	tcctaatgag	420
cttctccaag	aatataattg	ggaattggca	gaccaaccac	aaaatttggg	ggaaattttg	480
atgcattgcc	aaacaactct	aaaatatgca	attaaaacag	ggcatcctag	atacttcaat	540
caacttttcta	ctggtttggg	tatggttggg	ttagcagcag	actggctgac	atcaacagca	600
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cccggtggcg	ccatatctaa	catgtatgcc	atgatgatcg	cacgcttta	gatgttccca	780
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agtcattttt	ctctcaagaa	gggagctgca	gccttaggga	ttggaacaga	cagcgtgatt	900
ctgattaaat	gtgatgagag	agggaaaatg	attccatctg	atcttgaaag	aaggattctt	960
gaagccaaac	agaaagggtt	tgttcctttc	ctcgtgagtg	ccacagctgg	aaccaccgtg	1020
tacggagcat	ttgacccct	cttagctgtc	gctgacattt	gcaaaaagta	taagatctgg	1080

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atgcatgtgg atgcagcttg ggggtgggga ttactgatgt cccgaaaaca caagtggaaa	1140
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cctttgcagt gctctgtctt cctgggttaga gaagagggat tgatgcagaa ttgcaaccaa	1260
atgcatgcct cctacctctt tcagcaagat aaacattatg acctgtccta tgacactgga	1320
gacaaggcct tacagtgcgg acgccacgtt gatgttttta aactatggct gatgtggagg	1380
gcaaagggga ctaccgggtt tgaagcgcat gttgataaat gtttggagtt ggcagagtat	1440
ttatacaaca tcataaaaaa ccgagaagga tatgagatgg tgtttgatgg gaagcctcag	1500
cacacaaatg tctgcttctg gtacattcct ccaagcttgc gtactctgga agacaatgaa	1560
gagagaatga gtcgcctctc gaaggtggct ccagtgatta aagccagaat gatggagtat	1620
ggaaccacaa tggtcagcta ccaacccttg ggagacaagg tcaatttctt ccgcattggtc	1680
atctcaaacc cagcggcaac tcaccaagac attgacttcc tgattgaaga aatagaacgc	1740
cttgacaag atttataa	1758

<210> SEQ ID NO 18

<211> LENGTH: 537

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 18

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cttcgagatc tccgagatgc cttcagcaga gtgaagactt tctttcaaat gaaggatcag	180
ctggacaact tggtgttaaa ggagtccttg ctggaggact ttaagggta cctgggttgc	240
caagccttgt ctgagatgat ccagttttac ctggaggagg tgatgcccc agetgagaac	300
caagaccag acatcaaggc gcatgtgaac tccctggggg agaacctgaa gacctcagg	360
ctgaggctac ggcgtgtgca tcgatttctt cctgtgaaa acaagagcaa ggccgtggag	420
caggtgaaga atgcctttaa taagctccaa gagaaaggca tctacaaagc catgagttag	480
tttgacatct tcatcaacta catagaagcc tacatgacaa tgaagatagc aaactga	537

<210> SEQ ID NO 19

<211> LENGTH: 3969

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 19

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tctgacaagc gttccattat gggaaatcaa tccttacttt ggaaatggaa aggcggctcc	180
agttttacgt tacataaaaa attaatcgtg ccgaccgata aggaagcatc taaagcatgg	240
ggacgcagta gtacgccggg gttctccttc tggctgtaca acgagaaacc tattgatggg	300
tacttgacca tcgacttcgg cgaataattg atttcaacta gcgaagcaca agctggcttt	360
aagggttaaa tggacttcac cgggtggcgt acggtcggag tcagccttaa caatgactta	420
gagaatcgcg aaatgacctt gaatgtacg aatacgtcca gcgatggaac ccaggatagc	480
atcggtcgta gtctgggagc aaaggtggac agtatccgct tcaaagctcc gagcaacgtt	540

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tcacaggggg	aaatttatat	cgaccgtatt	atgttcagtg	tcgacgacgc	ccgttatcaa	600
tggagtgact	atcaagtaaa	gacccgtctg	tcagaacctg	agatccagtt	tcataatgtg	660
aagccgcagt	tgcccgtaac	gcccgaaaat	ttggcagcta	ttgatcttat	ccgccagcgc	720
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tcacaagata	aacagttatt	tgacaattat	gtcatcttag	gtaactacac	gaccttgatg	960
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acgtttacat cttacttcgg catcccacaa gaaattaagt tgtccccctt gccgggatcc	3000
ggcgccggca gccggcgag cgccggcagc ggccggcatgg cgcgcagtaa acagctggaa	3060
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cactgtgttc agtacaatca tcgggttaaa aaactcaatg aaatcagcaa actgggaatt	3900
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caccactga	3969

<210> SEQ ID NO 20

<211> LENGTH: 1224

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 20

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gtcattgtgg tgacctctgt ggtgggcaac gtggtagtga tgtggatcat cttagccac	180
aaaagaatga ggacagtgc gaactatctt ctggtgaacc tggccttcgc ggaggcctcc	240
atggctgcat tcaatacagt ggtgaacttc acctatgctg tccacaacga atgggtactac	300
ggcctgttct actgcaagtt ccacaacttc tttcccatcg ccgtgtctt cgcagctatc	360
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cggctgtcag ccacagccac caaagtggc atctgtgtca tctgggtcct ggctctcctg	480
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actgtgtgta tctacttctt cccctgctg gtgattggct atgcatacac cgtagtggga	660
atcacactat gggccagtga gatccccggg gactcctctg accgctacca cgagcaagtc	720
tctgccaagc gcaaggtggt caaaatgatg attgtcgtgg tgtgcacctt cgcctctg	780
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cggtgctgcc ccttcacag cgccggcgac tatgaggggc tggaaatgaa atccaccg	1020
tatctccaga cccagggcag tgtgtacaaa gtcagccgcc tggagaccac catctccaca	1080
gtggtggggg cccacgagga ggagccagag gacggcccca aggccacacc ctgctccctg	1140
gacctgacct ccaactgctc ttcacgaagt gactccaaga ccatgacaga gagcttcagc	1200
ttctcctcca atgtgctctc ctag	1224

<210> SEQ ID NO 21

<211> LENGTH: 762

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 21

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catccagcgg ctgcacctgt gtcctgcag tgcgggtca gcatgtgtcc agcgcgcagc	120
ctcctccttg tggtaccct ggtcctcctg gaccacctca gtttggccag aaacctcccc	180
gtggccactc cagaccagga aatgttccca tgccttcacc actccaaaa cctgctgagg	240
gccgtcagca acatgtcca gaaggccaga caaactctag aattttacc ttgcacttct	300
gaagagattg atcatgaaga tatcacaaaa gataaaacca gcacagtga ggctgttta	360
ccattggaat taaccaagaa tgagagttgc cttaaattcca gagagacctc ttccataact	420
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atggatccta agaggcagat ctttctagat caaaacatgc tggcagttat tgatgagctg	600
atgcaggccc tgaatttcaa cagtgagact gtgccacaaa aatcctccct tgaagaaccg	660
gatttttata aaactaaat caagctctgc atacttctc atgctttcag aattcgggca	720
gtgactattg atagagtgat gagctatctg aatgcttct aa	762

<210> SEQ ID NO 22

<211> LENGTH: 1083

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 22

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gggccagggg ggccggagcc gggctgggtt gatcctcgga cctggctaag cttccaaggc	120
cctcctggag ggccaggaat cgggcccggg gttgggccc gctctgaggt gtgggggatt	180
cccccatgcc ccccgccgta tgagttctgt ggggggatgg cgtactgtgg gccccagggt	240
ggagtggggc tagtgcccca aggcggcttg gagacctc agcctgaggg cgaagcagga	300
gtcggggtgg agagcaactc cgatggggcc tccccggagc cctgcacct caccctggt	360
gccgtgaagc tggagaagga gaagctggag caaaacccgg aggagtccca ggacatcaaa	420
gctctgcaga aagaactcga gcaatttgcc aagctcctga agcagaagag gatcacctg	480
ggatatacac aggcgatgt ggggctcacc ctgggggttc tatttgggaa ggtattcagc	540
caaacgacca tctgccgctt tgaggctctg cagcttagct tcaagaacat gtgtaagctg	600

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cggcccttgc tgcagaagtg ggtggaggaa gctgacaaca atgaaaatct tcaggagata	660
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gtgagaggca acctggagaa ttgttctctg cagtgccga aacccacact gcagcagatc	780
agccacatcg cccagcagct tgggctcgag aaggatgtgg tccgagtgtg gttctgtaac	840
cggcgccaga agggcaagcg atcaagcagc gactatgcac aacgagagga ttttgaggct	900
gctgggtctc ctttctcagg gggaccagtg tcctttcttc tggccccagg gccccathtt	960
ggtaccccag gctatgggag ccctcacttc actgcactgt actcctcggt ccctttccct	1020
gagggggaag cctttccccc tgtctccgtc accactctgg gctctcccat gcattcaaac	1080
tga	1083

<210> SEQ ID NO 23
 <211> LENGTH: 952
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 23

atgtacaaca tgatggagac ggagctgaag ccgcccggcc cgcagcaaac ttcggggggc	60
ggcgggcgga actccacgcg ggcggcggcc ggcggcaacc agaaaaacag cccggaccgc	120
gtcaagcggc ccatgaatgc ctctcatggtg tggccccgcg ggcagcggcg caagatggcc	180
caggagaacc ccaagatgca caactcggag atcagcaagc gcctgggcgc cagtggaata	240
cttttgtcgg agacggagaa ggcggcgttc atcgacgagg ctaagcggct gcgagcgtg	300
cacatgaagg agcaccggga ttataatac cggccccgcg ggaaaaccaa gacgctcatg	360
aagaaggata agtacacgct gcccgcgccg ctgctggccc ccggcggcaa tagcatggcg	420
agcgggggtcg ggggtggcgc cggcctgggc gcgggcggtg accagcgcac ggacagttac	480
gcgcacatga acggctggag caacggcagc tacagcatga tgcaggacca gctgggctac	540
ccgcagcacc cgggcctcaa tgcgcacggc gcagcgcaga tgcagcccat gcaccgctac	600
gacgtgagcg ccctgcagta caactccatg accagctcgc agacctacat gaacggctcg	660
cccacctaca gcatgtccta ctgcagcag ggcacccctg gcatggctct tggtcccatg	720
ggttcggtgg tcaagtccga ggccagctcc agccccctg tggttacctc ttctcccac	780
tccagggcgc cctgccaggc cggggacctc cgggacatga tcagcatgta tctccccgc	840
gccgaggtgc cggaaaccgc cgcgccagc agacttcaca tgtcccagca ctaccagagc	900
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<210> SEQ ID NO 24
 <211> LENGTH: 1374
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 24

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cgcgacgcgg cgctcttccc cggcgaggag ctgaagcacg cgcaccaacc cccgcaggcg	180
cagccccgcg ccgcgcaggc cccgcagccg gccagccgc ccgcacccg cccgcggctg	240
cctccagagg acctgggtcca gacaagatgt gaaatggaga agtatctgac acctcagctt	300

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cctccagttc ctataattcc agagcataaa aagtatagac gagacagtgc ctacgtcgta	360
gaccagttct tcaactgacac tgaagggtta ccttacagta tcaacatgaa cgtcttctct	420
cttgacatca ctcaactgag aactggcctc tacaaatccc agagaccgtg cgtaacacac	480
atcaagacag aacctgtgtg cattttcagc caccagagtgc aaacgactgc ccctcctccg	540
gccccgaccc aggcctctcc tgagttcacc agtatattca gctcacacca gaccgcagct	600
ccagaggtga acaatatttt catcaaaaaa gaacttccta caccagatct tcatctttct	660
gtccctaccc agcagggcca cctgtaccag ctactgaata caccgatct agatatgcc	720
agttctacaa atcagacagc agcaatggac actcttaatg tttctatgtc agctgccatg	780
gcaggcctta acacacacac ctctgtgtgt ccgcagactg cagtgaaca attccagggc	840
atgccccctt gcacatacac aatgccagt cagtttcttc cacaacaggc cacttacttt	900
cccccgctac caccaagctc agagcctgga agtccagata gacaagcaga gatgtccag	960
aatttaaccc cactctcatc ctatgtgtgt acaattgctt ctaaaactggc aattcacaat	1020
ccaaatttac ccaccacct gccagttaac tcacaaaaa tccaacctgt cagatacaat	1080
agaaggagta accccgattt ggagaaacga cgcattccact actgcgatta ccctggttgc	1140
acaaaagttt ataccaagtc ttctcattta aaagctcacc tgaggactca cactggtgaa	1200
aagccatata agtgtacctg ggaaggctgc gactggaggt tcgctcgatc ggatgagctg	1260
accgcctact accggaagca cacaggcgcc aagcccttcc agtgcggggt gtgcaaccgc	1320
agcttctcgc gctctgacca cctggccttg catatgaaga ggcaccagaa ctga	1374

<210> SEQ ID NO 25

<211> LENGTH: 918

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 25

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tcttctgtct agatgcctca cacggagact gtctctcttc tccctctctc catggatctg	180
cttattcagg acagccctga ttcttccacc agtcccaag gcaaacaccc cacttctgca	240
gagaatagtg tcgcaaaaaa ggaagacaag gtcccagtca agaaacagaa gaccagaact	300
gtgttctctt ccaccagct gtgtgtactc aatgatagat ttcagagaca gaaatacctc	360
agcctccagc agatgcaaga actctccaac atcctgaacc tcagctacaa acaggtgaag	420
acctggttcc agaaccagag aatgaaatct aagagggtggc agaaaaacaa ctggccgaag	480
aatagcaatg gtgtgacgca gaaggcctca gcacctacct accccagcct ctactcttcc	540
taccaccagg gatgcctggt gaacccgact gggaaccttc caatgtggag caaccagacc	600
tggacaatt caacctggag caaccagacc cagaacatcc agtcctggag caaccactcc	660
tggaaactc agacctggtg caccatcc ttggaacaatc aggcctggaa cagtcccttc	720
tataactgtg gagaggaatc tctgcagtcc tgcattgcagt tccagccaaa ttctctgcc	780
agtgacttgg aggtctgttt ggaagctgct ggggaaggcc ttaatgtaac acagcagacc	840
actaggtatt ttagtactcc acaaaccatg gatttattcc taaactactc catgaacatg	900
caacctgaag acgtgtga	918

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<210> SEQ ID NO 26
 <211> LENGTH: 630
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 26

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cccgaggagg cgccggagga cgccggcccg cgccgggacg agcctcagct gctgcacggt      120
gcgggcatct gtaagtgggt caacgtgcgc atggggttcg gcttctctgt catgaccgcc      180
cgccgggggg tcgcgctcga cccccagtg gatgtctttg tgcaccagag taagctgcac      240
atggaagggt tccggagctt gaaggagggt gaggcagtg agttcacctt taagaagtca      300
gccaaagggtc tggaatccat ccgtgtcacc ggacctggtg gagtattctg tattgggagt      360
gagaggcgcc caaaaggaaa gagcatgcag aagcgcagat caaaaggaga cagggtgtac      420
aactgtggag gtctagatca tcatgccaa gaaatgcaag tgccacccca gcccagaag      480
tgccacttct gccagagcat cagccatatg gtagcctcat gtccgctgaa ggcccagcag      540
ggccctagtg cacagggaag gccaacctac ttctgagagg aagaagaaga aatccacagc      600
cctaccctgc tcccggaggc acagaattga                                     630

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<210> SEQ ID NO 27
 <211> LENGTH: 1863
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 27

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caccagagcc tggacctccc gggccgaagc ctgccacccc ctgcgccttc ctgtacctt      120
ctgggcagcg aaccagctc tggcctgggc ctccagcccg agaccacct ccccgagggc      180
agcctgaagc ggtgctgcgt cttgggccta cccccacct cccagcctc ctccctaccc      240
tgtgcctcct ccgagctcac ctccatcac cgtctctccc agacgtctct ggtaacctgt      300
gtaaatggac tccggagccc cctcttgacg ggagatctgg ggggcccttc caagcggggc      360
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aaggcgagct tcctgaagca ggaaccgcg gatgagtttt cagagctctt tgggcctcac      480
cagcagggcc tgccgcccc ctatcccctg tctcagttgc cgctggccc aagccttgga      540
ggcctggggc tgggcctggc aggcagggtg gtggccgggc ggcaggcgtg ccgctgggtg      600
gactgctgtg cagcctatga gcagcaggag gagctggtgc ggcacatcga gaagagccac      660
atcgaccagc gcaaggcgca ggacttcacc tgcttctggg ctggtgcgt gcgccgtac      720
aagcccttca acgcccgtc caagctgctc atccacatgc gactgcactc gggcgagaag      780
cccaacaagt gcattgttga aggtgcagc aaggccttct cagggtgga gaacctcaag      840
atccacctga ggagccacac gggcgagaag ccgtacctgt gccagacccc gggttgccag      900
aaggccttca gcaactccag cgaccgcgcc aagcaccagc gcacccacct agacacgaag      960
ccgtacgcct gtcagatccc tggtgctcc aagcgtaca cagaccccag ctccctccgc      1020
aagcacgtca agggccattc agccaaagag cagcagggtc gtaagaagct gcatgcgggc      1080
cctgacaccg agggcgagct cctgaccgag tgtctggtcc tgcagcagct ccacacgtcc      1140

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acacagctgg ctgccagcga cggcaagggt ggcgtgtggc tgggccagga gctgctccca	1200
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gcgcacgacg taccttcag gcaccaccg ctggatgcca ccaccagttc ccaccaccat	1320
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cattctccgg ggggccagcc ctccccaca ctccccagca agcgtccta cccaccctc	1500
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gcgctgccac agcagccatc tgaagatgtg gtgtccagcg gcccagga ctgtggcttc	1800
ttcccattg gagccttga cactgctg ggcacatcc cctccatcta cacagacacc	1860
tga	1863

<210> SEQ ID NO 28

<211> LENGTH: 711

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 28

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gcagccgccg cagcggcagc gcagagcgcg cagcagcagc agcagcagca gcagcagcag	180
cagcagggcg cgcagctgag accggcggcc gacggccagc cctcaggggg cggtcacaa	240
tcagcgccca agcaagtcaa gcgacagcgc tcgtcttcgc ccgaactgat gcgctgcaaa	300
cgcgggtcct acttcagcgg ctttggttac agcctgccg agcagcagcc ggccgccgtg	360
gcgcgccgca acgagcgcga gcgcaaccgc gtcaagtggg tcaacctggg ctttgccacc	420
cttcgggagc acgtcccaaa cgcgcgggcc aacaagaaga tgagtaaggt ggagacactg	480
cgtcggcgcg tcgagtacat ccgcgcgtg cagcagctgc tggacgagca tgacgcgggtg	540
agcgcgcctt tccaggcagg cgtcctgtcg cccaccatct cccccaacta ctccaacgac	600
ttgaactcca tggccggctc gccggtctca tcctactcgt cggacgaggg ctcttaacgac	660
ccgctcagcc ccgaggagca ggagcttctc gacttcacca actgggtctg a	711

<210> SEQ ID NO 29

<211> LENGTH: 1149

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 29

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gtcatcttgg acaggtttct gctgcggctc aacgacagct tctggcatga gcagtgcgtg	180
cagtgcgcct cctgcaaaga gccctggag accacctgct tctaccggga caagaagctg	240
tactgcaagt atgactacga gaagctgtt gctgttaaat gtgggggctg cttcagggcc	300

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atcgctccca atgagtttgt tatgcgggcc cagaagagtg tataccacct gagctgcttc	360
tgctgctgtg tctgcgagcg acagcttcag aagggtgatg agtttgctct gaaggagggg	420
cagctgctct gcaaagggga ctatgagaag gagcgggagc tgctcagcct ggtgagccca	480
gcagcctcag actcaggtaa aagtgatgat gaagaaagtc tctgcaagtc agcccatggg	540
gcagggaaaag gaactgtgta ggaaggcaag gaccataagc gccccaaacg tccgagaacc	600
atcttgacaa ctcaacagag gcgagcattc aaggcctcat ttgaagtatc ctccaagccc	660
tgcaggaagg tgagagagac tctggctgca gagacagggc tgagtgtccg tgctgtccag	720
gtgtggttcc aaaaccagag agcgaagatg aagaagctgg ccaggcgaca gcagcagcag	780
cagcaagatc agcagaacac ccagaggctg agctctgctc agacaaacgg tgggtgggagt	840
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atgcctggag accacatgca cccttatggt gccgagcccc ttttccatga cctggatagc	1020
gacgacacct ccctcagtaa cctgggtgat tgtttcctag caacctcaga agctgggcct	1080
ctgcagtcca gagtgggaaa ccccatgac catctgtact ccatgcagaa ttcttacttc	1140
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<210> SEQ ID NO 30

<211> LENGTH: 3561

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 30

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catgtcagtg gcaaatatgc aagacacaga agtgtatatg gttgtccctt ggcgaaaaaa	180
agaaaaacac aagataaaca gccccaggaa cctgctccta aacgaaagcc atttgccgtg	240
aaagcagaca gctcctcagt ggatgagtgt gacgacagtg atgggactga ggacatggat	300
gagaaggagg aggatgaggg ggaggagtac tccgaggaca atgatgagcc aggggatgag	360
gacgaggagg acgaggagg ggaccgggag gaggaggagg agatcgagga ggaggatgag	420
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gaggaggaag aggaagaaga aaacgaagac catcaaatga attgtcacia tactcgaata	540
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gtggccaagt cattgttaaa cctcggaaca atcgctgagg atgcagccta ccgggccagg	660
actgagtcag aaatgaacag caatacctcc aatagtctgg aagacgatag tgacaaaaac	720
gaaaacctgg gtcggaagag tgagttgagt ttagacttag acagtgatgt tgtagagaa	780
acagtggact cccttaaaact attagcccaa ggacacgggt ttgtgctctc agaaaacatg	840
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ctcagtgaga ccaacccgca ggagaggaat ccgcagcaga acatgaacat ccgtcagcat	1080
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aacctcatgc	ggctggagga	gcagttgagc	ccccggtcga	gagtgtttgc	cagctgtgcg	1200
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gaaacggaaa	gagcaaaggc	catgaggag	aagatggcca	tggaaactgg	gaggaggagc	1380
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ggccaggggc	acatcactgg	gaagtacgcg	tcccatcgca	gcgcctccgg	gtgccccttg	2940
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cagatggaag	ccgatatgat	taaacacaga	actcagatta	ccacgatgga	gagcaacctg	3300
aagaccatcg	aagaggagaa	caaagtgatt	gagcagcaga	acgagtctct	cctccacgag	3360
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gtgagaggaa ttcagggtctg a	3561

<210> SEQ ID NO 31
 <211> LENGTH: 1332
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 31

atggcgaccg cagcgtctaa ccactacagc ctgctcacct ccagcgctc catcgtgcac	60
gccgagccgc ccggcgccat gcagcagggc gcggggggct accgcgaagc gcagagcctg	120
gtgcagggcg actacggcgc tctgcagagc aacggacacc cgctcagcca cgctcaccag	180
tggatcaccg cgctgtccca cggcggcggc ggcgggggcg gtggcggcgg cggggggggc	240
ggggggcgcg gcggggggcg cggcgacggc tccccgtggt ccaccagccc cctggggccag	300
ccggacatca agccctcggg ggtggtgcag cagggcggcc gcggagacga gctgcacggg	360
ccaggcgccc tgcagcagca gcatcagcag cagcaacagc aacagcagca gcaacagcag	420
caacagcagc agcagcagca gcaacagcgg ccgcccgcac tgggtgcacca cgcgcctaac	480
caccaccccg gaccgggggc atggcggagc gcggcggtcg cagcgacact cccaccctcc	540
atgggagcgt ccaacggcgg cttgctctac tcgcagccca gcttcacggt gaacggcatg	600
ctgggcgcgg gcgggcagcc ggccgggtcg caccaccacg gcctgcggga cgcgcacgac	660
gagccacacc atgcgcacca ccaccgcac ccgcactcgc acccacacca gcagccgccc	720
ccccgcggc ccccgaggg tccgcctggc caccaggcg cgcaccacga cccgcactcg	780
gacgaggaca gcgcgacctc ggacgacctg gagcagttcg ccaagcagtt caagcagcgg	840
cggatcaaac tgggatttac ccaagcggac gtggggctgg ctctgggcac cctgtatggc	900
aacgtgttct cgcagaccac catctgcagg tttgagggcc tgcagctgag cttcaagaac	960
atgtgcaagc tgaagccttt gttgaacaag tgggtggagg aggcggactc gtcctcgggc	1020
agccccacga gcatagacaa gatcgacgcg caagggcgca agcggaaaaa gcggacctcc	1080
atcgaggatg gcgtcaaggg ggctctggag agccatttcc tcaaatgccc caagccctcg	1140
gcccaggaga tcacctccct cgcggacagc ttacagctgg agaaggaggt ggtgagagtt	1200
tggttttgta acaggagaca gaaagagaaa aggatgaccc ctcccgagg gactctgccg	1260
ggcgccgagg atgtgtacgg ggggagtagg gacactccac cacaccacgg ggtgcagacg	1320
cccgccagtg ga	1332

<210> SEQ ID NO 32
 <211> LENGTH: 1071
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 32

atgaccaaata cgtacagcga gactgggctg atgggcgagc ctacgcccc aggtcctcca	60
agctggacag acgagtgtct cagttctcag gacgaggagc acgaggcaga caagaaggag	120
gacgacctcg aagccatgaa cgcagaggag gactcactga ggaacggggg agaggaggag	180
gacgaagatg aggacctgga agaggaggaa gaagagggaag aggaggatga cgatcaaaag	240

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cccaagagac gcgccccca aaagaagaag atgactaagg ctcgcctgga gcgtttttaa	300
ttgagacgca tgaaggctaa cgcccgagg cggaaccgca tgcacggact gaacgcggcg	360
ctagacaacc tgcgaaggt ggtgccttgc tattctaaga cgcagaagct gtccaaaatc	420
gagactctgc gcttggccaa gaactacatc tgggctctgt cggagatcct gcgctcaggc	480
aaaagcccag acctgggtctc ctctgttcag acgctttgca agggcttacc ccaaccacc	540
accaacctgg ttgcgggtg cctgcaactc aatcctcgga cttttctgcc tgagcagaac	600
caggacatgc cccccacct gccgacggcc agcgcttctc tcctgtaca cccctactcc	660
taccagtctc ctgggtgcc cagtccgcct tacggtacca tggacagctc ccatgtcttc	720
cacgttaagc ctccgccga cgctacagc gcagcgctgg agccctctt tgaaagccct	780
ctgactgatt gcaccagccc ttcctttgat ggacccctca gcccgccgct cagcatcaat	840
ggcaacttct ctttcaaaac cgaaccgtcc gccgagtttg agaaaaatta tgcctttacc	900
atgcactatc ctgcagcgac actggcaggg gcccaaagcc acggatcaat cttctcaggc	960
accgctgccc ctcgctgcga gatccccata gacaatatta tgtccttcga tagccattca	1020
catcatgagc gagtcagtag tgcacagctc aatgccatat tcatgatta g	1071

<210> SEQ ID NO 33

<211> LENGTH: 1269

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 33

atgcagaaca gtcacagcgg agtgaatcag ctcggtggtg tctttgtcaa cgggcggcca	60
ctgccggact ccaccggca gaagattgta gagctagctc acagcggggc cggccgtgc	120
gacatttccc gaattctgca ggtgtccaac ggatgtgtga gtaaaattct gggcaggat	180
tacgagactg gctccatcag acccagggca atcggtggta gtaaacgag agtagcgact	240
ccagaagttg taagcaaaat agcccagtat aagcgggagt gcccgctcat ctttgcttgg	300
gaaatccgag acagattact gtccgagggg gtctgtacca acgataacat accaagcgtg	360
tcatcaataa acagagttct tcgcaacctg gctagcgaaa agcaacagat gggcgagac	420
ggcatgtatg ataaactaag gatgtgaac gggcagaccg gaagctgggg caccgcct	480
ggttggtatc cggggacttc ggtgccaggg caacctacgc aagatggctg ccagcaacag	540
gaaggagggg gagagaatac caactccatc agttccaacg gagaagattc agatgaggct	600
caaatgagc ttcagctgaa gcggaagctg caaagaaata gaacatcctt taccgaagag	660
caaatgagg ccctggagaa agagtttgag agaaccatt atccagatgt gtttgccga	720
gaaagactag cagccaaaat agatctacct gaagcaagaa tacaggtagt gttttctaat	780
cgaagggcca aatggagaag agaagaaaa ctgaggaatc agagaagaca ggccagcaac	840
acacctagtc atattcctat cagcagtagt ttcagcacca gtgtctacca accaattcca	900
caaccacca caccggttcc ctcttcaca tctggctcca tgttgggccc aacagacaca	960
gccctcaca acacctacag cgctctgccg cctatgccca gcttcaccat ggcaataac	1020
ctgcctatgc aacccccagt cccagccag acctcctcat actcctgcat gctgcccacc	1080
agcccttcgg tgaatggggc gagttatgat acctacacc cccacatat gcagacacac	1140
atgaacagtc agccaatggg cacctcgggc accacttcaa caggactcat tccccgtgt	1200

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gtgtcagttc cagttcaagt tcccgaagt gaacctgata tgtctcaata ctggccaaga 1260

ttacagtaa 1269

<210> SEQ ID NO 34

<211> LENGTH: 1041

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 34

atgcacctgc cgggctgcgc gccagccatg gccagcggga gcttctcgt tgccggccac 60

ctgtcccgca gcccgggcgg gagcacctcg cgacttcaca gcatcgaggc catcctgggg 120

tttaccagg acgacgggat cctcggcacc ttccggcgg agcggggcgc cggggcgcg 180

aaggagcggg ataggaggct gggcgcgcg cccgcctgcc ccaaggcgcc cgaggaggc 240

tccgagccct ccccgccgc agccccggcg cccgcccccg agtacgaagc ccctcgacct 300

tactgcccc aggagcccg ggaggcacgg ccgagcccag ggctgcccgt cgggcccagc 360

accggcgaag cgaaactgtc agaggaggaa cagcccaga aaaagcatcg gcggaaccgc 420

acgactttca ccagctacca gctgcatgag ctggagcgcg cgttcgagaa gtccactac 480

ccggacgtgt acagccgcga ggagctggcc ggcaaggta acctaccaga ggtccgggtc 540

cagggtgtgt tccagaaccg acgggctaag tggcggcggc aggagaagct ggaagtgtcc 600

tccatgaagc tgcaggactc gccctctctc tcttcagcc gctccccgc ctcgcgacg 660

ctgtcgcccc tcggggcggg cccgggcagc ggtggcgggc cggctggggg cgcgtgccg 720

ctggagtctt ggctcgggccc gccgtgcgcg ggcgggggcg ccacggcgct gcagagcctg 780

ccgggcttcc ggccgcggc gcagagcctg cctgccagct acacgccacc gccgcgcct 840

ccgccttcc tgaactcccc gccgttggc cccggcctgc aacctctgc gccgcgcgcg 900

ccctcctacc cgtgcgggccc cggttcggg gacaagtcc cgctggacga ggcggaaccg 960

cgcaacagca gcatcgcggc gctgcgtctg aaagccaagg agcacatcca ggccatcggg 1020

aagccgtggc aggcctctc g 1041

<210> SEQ ID NO 35

<211> LENGTH: 900

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 35

atgatggcgt atatgaaccc ggggccccac tattctgtca acgccttggc cctaagtggc 60

cccagtgtgg atctgatgca ccaggctgtg cctaccaca gcgccccag gaagcagcgg 120

cgggagcgca ccaccttcac ccggagccaa ctggaggagc tggaggcact gtttgccaag 180

accagttacc cagacgtcta tccccgtgag gaggtggctc tgaagatcaa tctgcctgag 240

tccagggttc aggtttggtt caagaaccgg agggctaagt gcaggcagca gcgacagcag 300

cagaaacagc agcagcagcc cccagggggc caggccaagg cccggcctgc caagagggaag 360

gctgggcagt cccaagacc ctccacagat gtgtgtccag accctctggg catctcagat 420

tcctacagtc cccctctgcc cggccctca ggctcccaa ccacggcagt ggccactgtg 480

tccatctgga gccacgctc agagtccct ttgctgagg cgcagcgggc tgggctgggtg 540

gcctcagggc cgtctctgac ctccgcccc tatgcatga cctacgccc ggctccgct 600

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ttctgctctt cccctccgc ctatgggtct ccgagctctt atttcagcgg cctagacccc	660
tacctttctc ccattgggtgcc ccagctaggg ggcccggctc ttagccccc ctctggcccc	720
tccgtgggac cttccctggc ccagtccccc acctccctat caggccagag ctatggcgcc	780
tacagccccg tggatagctt ggaattcaag gacccacgg gcacctggaa attcacctac	840
aatcccatgg accctctgga ctacaaggat cagagtgcct ggaagtttca gatctttag	900

<210> SEQ ID NO 36

<211> LENGTH: 1563

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 36

atgcagtcgc aatcggggat cgtgccggat ttcgaagtcg gggaggagtt tcatgaagag	60
cccaaacctt attacgaact caaaagtcaa ccgctgaaga gcagcagttc cgccgagcat	120
cctggggcct ccaagcctcc gataagctcc tccagtatga catcacgcac cttgtctacgc	180
cagcaactca tgcgtgagca gatgcaggag caggagcgca gggagcagca gcagaagctg	240
caggcgggcc agttcatgca acagagagtg cccgtgagtc agacaccagc cataaacgtc	300
agtgtgcccc ccacccttcc ctctgccacg cagggtgccg tggaagtctc taagggtcag	360
accacactcg aaaacccccc caagtaccac atacagcaag cccaacggca gcaggtaaag	420
cagtaccttt ctaccacttt agcaataaaa catgccaaac aagtccctgag cttgccatgt	480
ccaaaccagc ctggcgatca tgtcatgcc aagggtgccg ggagcagcgc acccaacagc	540
cccatggcta tgcttacgct taactccaac tgtgaaaaag agggatttta taagtttgaa	600
gagcaaaaac gggcagagag cgagtgccca ggcatgaaca cacattcacg agcgtcctgt	660
atgcagatgg atgatgtaat cgatgacatc attagcctag aatcaagtta taatgaggaa	720
atcttgggct tgatggatcc tgctttgcaa atggcaata cgttgccctgt ctcggaac	780
ttgattgac tttatggaaa ccaaggtctg cccccaccag gcctcaccat cagcaactcc	840
tgtccagcca accttcccaa cataaaaagg gagctcacag agtctgaagc aagagcactg	900
gccaaagaga ggcagaaaaa ggacaatcac aacctgattg aacgaagaag aagatttaac	960
ataaatgacc gcattaaaga actagggtact ttgattccca agtcaaatga tccagacatg	1020
cgctggaaca agggaaacct cttaaaagca tccgtggact atatccgaaa gttgcaacga	1080
gaacagcaac gcgcaaaaga acttgaaaac cgacagaaga aactggagca cgccaaccgg	1140
catttgttgc tcagaataca ggaacttgaa atgcaggctc gagctcatgg actttccctt	1200
attccatcca cgggtctctg ctctccagat ttggtgaatc ggatcatcaa gcaagaaccc	1260
gttcttgaga actgcagcca agacctcctt cagcatcatg cagacctaac ctgtacaaca	1320
actctcgatc tcacggatgg caccatcacc ttcaacaaca acctcggaac tgggactgag	1380
gccaaaccaag cctatagtgt ccccaaaaa atgggatcca aactggaaga catcctgatg	1440
gacgacaccc tttctcccg tgggtgcact gatccactcc tttcctcagt gtcccccgga	1500
gcttccaaaa caagcagccg gaggagcagt atgagcatgg aagagacgga gcacacttgt	1560
tag	1563

<210> SEQ ID NO 37

<211> LENGTH: 894

<212> TYPE: DNA

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<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 37

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atgatgtctt atcttaagca accgccttac gcagtcaatg ggctgagtct gaccacttcg      60
ggatatggact tgctgcaccc ctccgtgggc taaccggggc cctgggcttc ttgtcccgca    120
gccaccccc ggaacacagcg ccgggagagg acgacgttca ctccggcgca gctagatgtg    180
ctggaagcac tgtttgccaa gacccggtag ccagacatct tcatgcgaga ggaggtggca    240
ctgaaaatca acttgcccca gtcgagggtg caggtatggt ttaagaatcg aagagctaag    300
tgccgccaac aacagcaaca acagcagaat ggaggtcaaa acaaagtga acctgccaaa    360
aagaagacat ctccagctcg ggaagtgaat tcagagagtg gaacaagtgg ccaattcact    420
ccccctcta gcacctcagt cccgaccatt gccagcagca gtgctcctgt gtctatctgg    480
agcccagctt ccattctccc actgtcagat cccttggtcca cctcctcttc ctgcatgcag    540
aggtcctatc ccattgaccta tactcaggct tcaggttata gtcaaggata tgctgggtca    600
acttcctact ttgggggcat ggactgtgga tcataattga cccctatgca tcaccagctt    660
cccgaccag gggccacact cagtcccatg ggtaccaatg cagtcaccag ccattcfaat    720
cagtcaccag cttctcttcc caccaggga tatggagctt caagcttggg ttttaactca    780
accactgatt gcttgatta taaggaccaa actgcctcct ggaagcttaa cttcaatgct    840
gactgcttgg attataaaga tcagacatcc tcgtggaaat tcaggtttt gtga          894

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<210> SEQ ID NO 38

<211> LENGTH: 714

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 38

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atggccctgc ccccagccc cctggccatg gaatatgtca atgaatttga cttgatgaag      60
tttaggtaa agcgggaacc ctctgagggc cgacctggcc cccctacagc ctcactgggc    120
tccacacctt acagctcagt gcctccttca cccaccttca gtgaaccagg catgggtggg    180
gcaaccgagg gcacccggcc aggcctggag gagctgtact ggctggctac cctgcagcag    240
cagctggggg ctggggaggc attggggctg agtcctgaag aggccatgga gctgctgcag    300
ggtcaggggc cagtcctgtg tgatggggcc catggctact acccaggag cccagaggag    360
acaggagccc agcacgtcca gctggcagag cgggttttccg acgcggcgct ggtctcgatg    420
tctgtgcggg agctaaaccg gcagctgcgg ggctgcgggc gcgacgaggc gctgcggctg    480
aagcagaggg gccgcacgct gaagaaccgc ggctacgcgc aggcctgtcg ctccaagcgg    540
ctgcagcagc ggcgcgggct ggaggccgag cgcgcccgcc tggccgcca gctggacgcg    600
ctgcggggcc aggtggcccg cctggcccgg gagcgcgacg tctacaaggc tcgctgtgac    660
cgggtaacct cgagcggccc cgggtccggg gacccctccc acctcttct ctga          714

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<210> SEQ ID NO 39

<211> LENGTH: 1440

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 39

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atgaggcagc cacctggcga gtctgacatg gctgtcagcg acgcgctgct cccatcttct      60

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tccacgttcg cgtctggccc ggcgggaagg gagaagacac tgcgtcaagc aggtgccccg	120
aataaccgct ggcgggagga gctctccccc atgaagcgac tccccccagt gcttccccgc	180
cggccctatg acctggcggc ggcgaccgtg gccacagacc tggagagcgg cgagagccgt	240
gcggcttgcg gcggtagcaa cctggcggcc ctacctcgga gagagaccga ggagtccaac	300
gatctcctgg acctggactt tattctctcc aattcgctga cccatcctcc ggagtcagtg	360
gcccgcaccg tgtcctcgtc agcgtcagcc tctctctcgt cgtcgccgtc gagcagcggc	420
cctgccagcg cggccctccac ctgcagcttc acctatccga tccgggcccgg gaacgacccg	480
ggcgtggcgc cggggcgacac gggcggaggc ctctctctatg gcagggagtc cgctccccct	540
ccgacggctc ccttcaacct ggcggacatc aacgacgtga gccctcggg cggttcgtg	600
gccgagctcc tgcggccaga attggaccgc gtgtacattc cggcgacgca gccgcagccg	660
ccaggtggcg ggctgatggg caagttcgtg ctgaaggcgt cgtcgagcgc ccttggcagc	720
gagtacggca gccctcggt catcagcgtc agcaaaggca gccctgacgg cagccaccgc	780
gtggtggtgg cggccctacaa cggcggggcg cgcgcacgt gcccgaagat caagcaggag	840
gcggtctctt cgtgcaccca cttggcgct ggacccccctc tcagcaatgg ccaccggccg	900
gtgcacacg acttccccct gggggcgag ctccccagca ggactacccc gacctgggt	960
cttgaggaag tgctgagcag cagggaactgt caccctgccc tgccgcttcc tcccggttc	1020
catccccacc cggggcccaa ttaccatcc ttctgcccc atcagatgca gccgcaagtc	1080
cgcgcgtcc attaccaaga gctcatgcca cccggttctc gcatgccaga ggagcccaag	1140
ccaaagaggg gaagacgac gtggcccccg aaaaggaccg cccccacac ttgtgattac	1200
gcgggctgcg gcaaaaccta cacaaagagt tcccatctca aggcacacct gcgaaccac	1260
acaggtgaga aaccttacca ctgtgactgg gacggctgtg gatggaaatt cggccgtca	1320
gatgaactga ccaggcacta ccgtaaacac acggggcacc gcccggtcca gtgcaaaaa	1380
tgcgaccgag cattttcag gtcggaccac ctgccttac acatgaagag gcatttttaa	1440

<210> SEQ ID NO 40

<211> LENGTH: 819

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 40

atgttcgtca aatccgagac cttggagttg aaggaggaag aggacgtgtt agtgctgctc	60
ggatcggcct ccccgccctt ggcggccctg acccgcgtgt catccagcgc cgacgaagaa	120
gaggaggagg agccgggcgc gtcaggcggg gcgcgtcggc agcgcggggc tgaggccggg	180
cagggggcgc ggggcggcgt ggctgcgggt gcggaggcgt gccggccgc acggctgctg	240
ggtctggtac acgattgcaa acggcgccct tcccgggcgc gggcgtctc ccgaggcgc	300
aagacggccg agacgggtga gcgcacaaag aagaccgta gactgaaggc caacaaccgc	360
gagcgaacc gcatgcacaa cctcaacgcg gcaactggacg cgctgcgcga ggtgctcccc	420
acgttccccg aggaagccaa gtcaccaag atcgagacc tgcgttcgc ccacaactac	480
atctgggcac tcaccgagac cctgcgctg gcggatcact gcggggcgcg cgcgggggc	540
ctgccggggg cgctcttctc cgaggcagtg ttgctgagcc cgggaggagc cagcgccgc	600
ctgagcagca gcggagacag cccctcgccc gcctccacgt ggagttgcac caacagcccc	660

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gcgcgcgtcct cctccgtgtc ctccaattcc acctccccct acagctgcac tttatcgccc	720
gccagccccg cgggtcaga catggactat tggcagcccc cacctccga caagcaccgc	780
tatgcacctc acctcccac agccagggat tgtatctag	819

<210> SEQ ID NO 41
 <211> LENGTH: 1326
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 41

atggtgcagc aggcggagag cttggaagcg gagagcaacc tgccccgga ggcgctggac	60
acggaggagg gcgaattcat ggcttgacg cgggtggccc tggacgagag cgaccagac	120
tggtgcaaga cggcgctcggg ccacatcaag cggccgatga acgcgttcat ggtatggtcc	180
aagatcgaac gcaggaagat catggagcag tctccggaca tgcacaacgc cgagatctcc	240
aagaggctcg gcaagcgtg gaaaatgctg aaggacagcg agaagatccc gttcatccgg	300
gaggcggagc ggctgcggct caagcacatg gccgactacc ccgactacaa gtaccggccc	360
cggaaaaagc ccaaatgga ccctcggcc aagcccagcg ccagccagag ccagagaag	420
agcgcggccg cgcggcgcg cgggagcgcg ggcggaggcg cggcggtgc caagacctcc	480
aagggtcca gcaagaaatg cggcaagctc aaggcccccg cggcgcggg cgccaaggcg	540
ggcgcgggca aggcggccca gtcggggac tacggggcg cggcgacga ctacgtgctg	600
ggcagcctgc gcgtgagcgg ctcgggcggc ggcggcgcg gcaagacggt caagtgcgtg	660
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caggagcccg acgaggagga cgaggaacca ccgcaccagc agctcctgca gccgcggggg	780
cagcagccgt cgcagctgct gagacgtac aacgtcgcca aagtgcctgc cagccctacg	840
ctgagcagct cggcggagtc ccccgaggga gcgagcctct acgacgaggt gcgggcccgc	900
gcgacctcgg cgcgcggggg cggcagccgc ctctactaca gcttcaagaa catcaccaag	960
cagcaccgcg cgcgcctcgc gcagcccgcg ctgtcgcccg cgtcctcgcg ctcggtgtcc	1020
acctcctcgt ccagcagcag cggcagcagc agcggcagca gcggcgagga cgccgacgac	1080
ctgatgttcg acctgagctt gaattttctt caaagcgcgc acagcgccag cgagcagcag	1140
ctggggggcg cgcggcgcg cgggaacctg tccctgtcgc tgggtggataa ggatttgat	1200
tcgttcagcg agggcagcct gggtcccccac ttcgagtcc ccgactactg cagcgggag	1260
ctgagcgaga tgatcgcggg ggactggctg gaggcgaact tctccgacct ggtgttcaca	1320
tattga	1326

<210> SEQ ID NO 42
 <211> LENGTH: 1050
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 42

atgggagaca tgggagatcc accaaaaaaa aaacgtctga tttccctatg tgttggttgc	60
ggcaatcaga ttacagatca gtatatctg agggtttctc cggatttgga atggcatgcg	120
gcatgtttga aatgtgcgga gtgtaacag tatttgagc agagctgtac atgctttgtt	180
agggatggga aaacctactg taaaagagat tatatcaggt tgtacgggat caaatgcgcc	240

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aagtgcagca tgggttcag caagaacgac ttcgtgatgc gtgcccgtc caaggtgtat	300
cacatcgagt gtttccgtg tgtggcctgc agccgccagc tcacccctgg ggacgaattt	360
gcgcttcggg aggaaggctc cttctgccga gcagaccagc atgtggtgga gagggccagt	420
ctaggcgctg gcgacccgct cagtccctg catccagcgc ggccactgca aatggcagcg	480
gagcccatct ccgcccaggca gccagccctg cggccccacg tcacacaagca gccggagaag	540
accacccgcg tgcggactgt gctgaacgag aagcagctgc acaccttgcg gacctgtac	600
gccgcaaacc cgcggccaga tgcgtcatg aaggagcaac tggtagagat gacgggctc	660
agtccccgtg tgatccgggt ctggtttcaa aacaagcggg gcaaggacaa gaagcgaagc	720
atcatgatga agcaactcca gcagcagcag cccaatgaca aaactaatat ccaggggatg	780
acaggaactc ccatggtggc tgccagtcca gagagacacg acggtggctt acaggctaac	840
ccagtggaag taaaaagta ccagccacct tggaaagtac tgagcgactt cgccttgca	900
agtgacatag atcagcctgc ttttcagcaa ctggtaatt tttcagaagg aggaccgggc	960
tctaattcca ctggcagtga agtagcatca atgtcctctc aacttcaga tacacctaac	1020
agcatggtag ccagtcctat tgaggcatga	1050

<210> SEQ ID NO 43

<211> LENGTH: 1194

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 43

atgtgtctgg aaacggggct cgagcgcgac cgagcgaggc ccggggccgc cgccgtctgc	60
accttgggcg ggactcggga gatcccgtg tgcgtggct gtgaccagca catcctggac	120
cgttcatcc tcaaggctct ggaccgccac tggcacagca agtgtctcaa gtgcagcgac	180
tgccacacgc cactggccga gcgtgcttc agccgagggg agagcgttta ctgcaaggac	240
gactttttca agcgttcgga gaccaagtgc gccgcgtgcc agctgggcat ccgcccacg	300
caggtggtgc gcccgcccca ggacttcgtg taccacctgc actgctttgc ctgcgtctg	360
tgcaagcggc agctggccac ggcgcagcag ttctacctca tggaggacag ccggctcgtg	420
tgcaaggcgg actacgaac cgccaagcag cgagaggccg aggccacggc caagcggccg	480
cgcacgacca tcaccgcaa gcagctggag acgtgaaga gcgttacaa cactcgccc	540
aagccggcgc gccacgtgcg cgagcagctc tcgtccgaga cgggcctgga catgcgctg	600
gtgcaggttt ggttcagaa ccgccgggccc aaggagaaga ggctgaagaa ggacgccggc	660
cggcagcgct gggggcagta tttccgaac atgaagcgt cccgcggcgg ctccaagtcg	720
gacaaggaca gcgttcagga ggggcaggac agcgacgctg aggtctcctt ccccgatgag	780
ccttccttgg cggaaatggg cccggccaat ggcctctacg ggagcttggg ggaaccacc	840
caggccttgg gccggccctc gggagccctg ggcaacttct ccctggagca tggaggcctg	900
gcaggcccgag agcagtaccg agagctgctt cccggcagcc cctacggtgt ccccccac	960
cccgccgccc cgcagagcct ccttgcccc cagccctcc tctccagcct ggtgtaccca	1020
gacaccagct tgggccttgt gccctcggga gcccccgcg gggccccacc catgagggtg	1080
ctggcaggga acggacccag tctgacctc tccacgggga gcagcggggg ttaccccgac	1140
ttccctgcca gccccgcctc ctggctggat gaggtagacc acgtcagtt ctga	1194

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<210> SEQ ID NO 44
 <211> LENGTH: 1350
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 44

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gaagtgcagc tgcctggaatc tggcggcgga ctggtgcagc ctggcggtc tctgagactg      60
tcttgtgccg cctccggcta caccttcacc acctactgga tcacctgggt ccgacaggct      120
cccggaagg gactggaatg ggtcggagac atctaccccg gctcctccat ctgcaactac      180
gccgacaagt tcaagggcgg gtccaccatc tccgtggaca actccaagtc caccgcctac      240
ctccagatga actcctgag agccgaggac accgcggtgt actactgcgc cagagaggac      300
ggctacgacg cttggtttgc ttactggggc cagggcaccg tggtcaccgt gtcacttgca      360
tccaccaagg gcccatcggt ctccccctg gcacctcct ccaagagcac ctctgggggc      420
acagcggccc tgggtgctct ggtcaaggac tacttcccg aaccggtgac ggtgtcgtgg      480
aactcaggcg cctgaccag cgcggtgcac accttcccg ctgtctaca gtcctcagga      540
ctctactccc tcagcagcgt ggtgaccgtg cctccagca gcttgggcac ccagacctac      600
atctgcaacg tgaatcaca gcccgcaac accaaggtgg acaagaaagt tgagccaaa      660
tcttgtgaca aaactcacac atgccaccg tgccagcac ctgaactcct ggggggaccg      720
tcagtcttcc tcttcccccc aaaaccaag gacacctca tgatctcccg gacctctgag      780
gtcacatgcg tgggtgtgga cgtgagccac gaagacctg aggtcaagtt caactggtac      840
gtggacggcg tggaggtgca taatgccaag acaaagccgc gggaggagca gtacaacagc      900
acgtaccgtg tggtcagcgt cctcaccgtc ctgcaccagg actggctgaa tggcaaggag      960
tacaagtgca aggtctccaa caaagccctc ccagcccca tcgagaaaac catctccaaa     1020
gccaaaggcg agccccgaga accacagggt tacacctgc ccccatcccg ggatgagctg     1080
accaagaacc aggtcagcct gacctgcctg gtcaaaggct tctatcccag cgacatcgcc     1140
gtggagtggg agagcaatgg gcagccggag aacaactaca agaccacgcc tcccgtgctg     1200
gactccgacg gctccttctt cttatattca aagctcaccg tggacaagag caggtggcag     1260
caggggaacg tcttctcatg ctccgtgatg catgaggctc tgcacaacca ctacacgcag     1320
aagagcctct cctgtctctc cgggaaatga                                     1350

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<210> SEQ ID NO 45
 <211> LENGTH: 396
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 45

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atgaagttgc ctgtaggct gttggtgctg atgttctgga ttcttggtc cagcagtgat      60
gttatgatga cccaaactcc actctccctg cctgtcagtc ttggagatca agcctccatc     120
tcttgcagat ctagtccag tcttgtacat agtaatggaa acacctatct agaatggtat     180
atgcagaaac caggccagtc tccaatgctc ctgatctaca aagtttccaa ccgattttct     240
gggttcccag acaggttcag tggcagtgga tcagggacag atttcacact caagatcagc     300
agcgtggagg ctgaggatct gggagttttt tactgcttcc aaggttcacg tgttccgctc     360
acgttcggtg ctgggaccaa gctggagctg aaacgg                                     396

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<210> SEQ ID NO 46
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 46

Arg Gln Ile Lys Ile Trp Phe Gln Asn Arg Arg
1 5 10

<210> SEQ ID NO 47
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 47

Tyr Gly Arg Lys Lys Arg Arg Gln Arg Arg Arg
1 5 10

<210> SEQ ID NO 48
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 48

Leu Leu Asn Phe Asp Leu Leu Lys Ile Leu Leu Ala Gly Asp Trp Glu
1 5 10 15

Ser Asn Pro Cys Pro
20

<210> SEQ ID NO 49
<211> LENGTH: 58
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 49

Met Phe Pro Thr Ile Pro Leu Ser Arg Leu Phe Asp Asn Ala Met Leu
1 5 10 15

Arg Ala His Arg Leu His Gln Leu Ala Phe Asp Thr Tyr Gln Glu Phe
20 25 30

Glu Glu Ala Tyr Ile Pro Lys Glu Gln Lys Tyr Ser Phe Leu Gln Asn
35 40 45

Pro Gln Thr Ser Leu Cys Phe Ser Glu Ser
50 55

<210> SEQ ID NO 50
<211> LENGTH: 58
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 50

Arg Thr Gly Gln Ile Phe Lys Gln Thr Tyr Ser Lys Phe Asp Thr Asn
1 5 10 15

Ser His Asn Asp Asp Ala Leu Leu Lys Asn Tyr Gly Leu Leu Tyr Cys
20 25 30

Phe Arg Lys Asp Met Asp Lys Val Glu Thr Phe Leu Arg Ile Val Gln
35 40 45

Cys Arg Ser Val Glu Gly Ser Cys Gly Phe
50 55

-continued

<210> SEQ ID NO 51
<211> LENGTH: 58
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 51

Ile Pro Thr Pro Ser Asn Arg Glu Thr Gln Gln Lys Ser Asn Leu Glu
1 5 10 15
Leu Leu Arg Ile Ser Leu Leu Ile Gln Ser Val Tyr Gly Ala Ser Asp
20 25 30
Ser Asn Val Tyr Asp Leu Leu Lys Asp Leu Glu Glu Gly Ile Gln Thr
35 40 45
Leu Met Gly Arg Leu Glu Asp Gly Ser Pro
50 55

<210> SEQ ID NO 52
<211> LENGTH: 134
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 52

Met Asp Met Arg Thr Pro Ala Gln Phe Leu Gly Ile Leu Leu Leu Trp
1 5 10 15
Phe Pro Gly Ile Lys Cys Asp Ile Lys Met Thr Gln Ser Pro Ser Ser
20 25 30
Met Tyr Ala Ser Leu Gly Glu Arg Val Thr Phe Thr Cys Lys Ala Ser
35 40 45
Gln Asp Ile Asn Asn Tyr Leu Cys Trp Phe Gln Gln Lys Pro Gly Lys
50 55 60
Ser Pro Lys Thr Leu Ile Tyr Arg Ala Asn Arg Leu Val Asp Gly Val
65 70 75 80
Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Gln Asp Tyr Ser Leu Thr
85 90 95
Ile Ser Ser Leu Glu Tyr Glu Asp Met Gly Ile Tyr Tyr Cys Leu Gln
100 105 110
Tyr Asp Glu Phe Pro Tyr Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile
115 120 125
Lys Arg Ala Asp Ala Ala
130

<210> SEQ ID NO 53
<211> LENGTH: 200
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 53

Met Ala Ala Ala Gly Gln Leu Cys Leu Leu Tyr Leu Ser Ala Gly Leu
1 5 10 15
Leu Ser Arg Leu Gly Ala Ala Phe Asn Leu Asp Thr Arg Glu Asp Asn
20 25 30
Val Ile Arg Lys Tyr Gly Asp Pro Gly Ser Leu Phe Gly Phe Ser Leu
35 40 45
Ala Met His Trp Gln Leu Gln Pro Glu Asp Lys Arg Leu Leu Leu Val
50 55 60
Gly Ala Pro Arg Ala Glu Ala Leu Pro Leu Gln Arg Ala Asn Arg Thr

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65					70					75					80				
Gly	Gly	Leu	Tyr	Ser	Cys	Asp	Ile	Thr	Ala	Arg	Gly	Pro	Cys	Thr	Arg				
				85					90					95					
Ile	Glu	Phe	Asp	Asn	Asp	Ala	Asp	Pro	Thr	Ser	Glu	Ser	Lys	Glu	Asp				
				100					105					110					
Gln	Trp	Met	Gly	Val	Thr	Val	Gln	Ser	Gln	Gly	Pro	Gly	Gly	Lys	Val				
				115					120					125					
Val	Thr	Cys	Ala	His	Arg	Tyr	Glu	Lys	Arg	Gln	His	Val	Asn	Thr	Lys				
				130					135					140					
Gln	Glu	Ser	Arg	Asp	Ile	Phe	Gly	Arg	Cys	Tyr	Val	Leu	Ser	Gln	Asn				
				145					150					155					
Leu	Arg	Ile	Glu	Asp	Asp	Met	Asp	Gly	Gly	Asp	Trp	Ser	Phe	Cys	Asp				
				165					170					175					
Gly	Arg	Leu	Arg	Gly	His	Glu	Lys	Phe	Gly	Ser	Cys	Gln	Gln	Gly	Val				
				180					185					190					
Ala	Ala	Thr	Phe	Thr	Lys	Asp	Phe												
				195					200										

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<210> SEQ ID NO 54
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
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<400> SEQUENCE: 54

[illegible]

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<210> SEQ ID NO 55
<211> LENGTH: 98
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
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<400> SEQUENCE: 55

Glu	Val	Gln	Leu	Gln	Gln	Ser	Gly	Pro	Glu	Leu	Val	Lys	Pro	Gly	Asp
1				5					10					15	
Ser	Val	Lys	Met	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asp	Tyr
			20					25					30		
Tyr	Met	Asp	Trp	Val	Lys	Gln	Ser	His	Gly	Lys	Ser	Leu	Glu	Trp	Ile
		35					40					45			
Gly	Tyr	Ile	Tyr	Pro	Asn	Asn	Gly	Gly	Thr	Ser	Tyr	Asn	Gln	Lys	Phe

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50	55	60	
Lys Gly Lys Ala Thr	Leu Thr Val Asp Lys Ser	Ser Ser Thr Ala Tyr	
65	70	75	80
Met Glu Leu His Ser	Leu Thr Ser Glu Asp Ser	Ala Val Tyr Tyr Cys	
	85	90	95

Ala Arg

<210> SEQ ID NO 56
 <211> LENGTH: 19
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 56

tccctatcag tgatagaga	19
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<210> SEQ ID NO 57
 <211> LENGTH: 260
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 57

ttgtcaggca acgtggcgtg gtgtgcactg tgtttgetga cgcaaccccc actggttggg	60
gcattgccac cacctgtcag ctcccttccg ggactttcgc tttccccctc cctattgcc	120
cggcggaact catcgccgcc tgccttgccc gctgctggac aggggctcgg ctgttgggca	180
ctgacaattc cgtgggtgtg tcgggggaagc tgacgtcctt tccatggctg ctgcctgtg	240
ttgccacctg gattctgcgc	260

<210> SEQ ID NO 58
 <211> LENGTH: 563
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 58

gagctcacgg ggacagcccc cccccaaagc ccccagggat gtaattacgt cctcccccg	60
ctagggggca gcagcgagcc gcccggggct ccgctccggt ccggcgctcc ccccgcatcc	120
ccgagccggc agcgtgctgg gacagcccg gacaggggaa ggtggcacgg gatcgctttc	180
ctctgaacgc ttctcgctgc tctttgagcc tgcagacacc tggggggata cggggaaaaa	240
gcttttaggt gaaagagaga tttagaatga caggcgcgcc tggccatata tcgatacgg	300
accgagttgg cgcgccctgg agctcacggg gacagcccc ccccaaagcc cccagggatg	360
taattacgtc cctccccgc tagggggcag cagcgagccg cccggggctc cgtccggtc	420
cggcgctccc cccgcatccc cgagccggca gcgtgcgggg acagcccggg cacggggaag	480
gtggcacggg atcgctttcc tctgaacgct tctcgctgct ctttgagcct gcagacacct	540
gggggggatac ggggaaaaag ctt	563

<210> SEQ ID NO 59
 <211> LENGTH: 1072
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 59

cacaggtgct aatatttcaa ttatagctct aaatcagtg ccaaaagcgt ggccccgaca	60
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aaaggcttct atgggcttga ttggagtagg cacagcatct gaagtctatc aaagtgtctac	120
aattttacat cgtttaagac cagatgggca ggaagtcacc atccaaccta tgattacttt	180
cattcccatt aattttagg ggagagattt gttacagcaa tggggtacag aaatcaccat	240
tccctctgca atttacagtc ctgagagcca aagaatgatg accaaaatgg gatataattcc	300
tggaagaggga ttaggaaaaa aatgaggacg gaataaccca gccgattgag gtctcagtta	360
aatttgaccg aagaggaatt ggatatacctt cttagggtgag gccactgttg agcctccaaa	420
gcccattgca ttaaaacgga aaactccaaa acctgtttgg gttgatcagt ggccgcttcc	480
tcaacaaaaa ctggaggcat tacattcatt ggcaaaagaa caattagaca aaggacatat	540
tgaaccatct ttttcacat ggaacgccc agtttttgta attcagaaaa aattcgctag	600
atggcgcata ttgaccgact tatgagcggg taacgcagcc attcaaccga tgggagctct	660
ccaaccaggg ttgccatctc cgcccatgat cccaaaagaa tggccgatgg taataattga	720
tctaaaagat tgtgtttttt actattcctt tggctcctca ggattttgag aaatttgcat	780
ttacaatacc tgccatgaat aacaaggaa cagctactat atacgatgg aaagtgtgac	840
ccaaggaat gttaaatagt caaactttt ttggaaagg tttcagctt gtcagggatc	900
agtttctga ttgctatatt attcattatg ttgatgacat tttatgtgct gctgcaagca	960
gggacagacg gattgagtgt tttgctaggc tacaagaagt ggtgggactt acaggctctg	1020
ctatagcacc agataaaatc catactacac cctatcatta cctgggaatg ag	1072

<210> SEQ ID NO 60

<211> LENGTH: 2177

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 60

gaaggtgatg cagtcgaagc cattgtggag gagtccgaaa cttttattaa aggaaaggag	60
agaaagactt accagagacg ccgggaaggg gccaggaag aagatgcctg ccacttacc	120
cagaaccaga cggatggggg tgaggtggc caggatgtca acagcagtgt acagatgggtg	180
atgatggaac agctggacc cacccttctt cagatgaaga ctgaagtaat ggagggcaca	240
gtggctccag aagcagaggc tgctgtggac gataccaga ttataacttt acaggttgta	300
aatatggagg aacagcccat aaacatagga gaacttcagc ttgttcaagt acctgttct	360
gtgactgtac ctgttctac cacttcagta gaagaacttc agggggctta tgaaatgaa	420
gtgtctaaag agggccttgc ggaaagtga cccatgatat gccacacct acctttgcct	480
gaaggggttc aggtggttaa agtgggggcc aatggagagg tggagacact agaacaagg	540
gaacttcac cccaggaaga tcctagtgg caaaaagacc cagactatca gccaccagcc	600
aaaaaaacaa agaaaaccaa aaagagcaaa ctgcgttata cagaggaggg caaagatgta	660
gatgtgtctg tctacgattt tgaggaagaa cagcaggagg gtctgttca gaggttaatg	720
cagagaaagt ggttggtaat atgaagctc caaagccaac aaaaattaaa aagaaaggtg	780
taaagaagac attccagtgt gagctttgca gttacacgtg tccacggcgt tcaaatttgg	840
atcgtcacat gaaaagccac actgatgaga gaccacaca gtgccatctc tgtggcagg	900

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cattcagaac agtcaccctc ctgaggaatc accttaacac acacacaggt actcgtcctc 960
acaagtgtccc agactgcgac atggcctttg tgaccagtgg agaattgggt cggcategtc 1020
gttacaacaa caccacagag aagccattca agtgttccat gtgcgattac gccagtgtag 1080
aagtcagcaa attaaaacgt cacattcgct ctcatactgg agagcgtccg tttcagtgca 1140
gtttgtgcag ttatgccagc agggacacat acaagctgaa aaggcactg agaaccatt 1200
caggggaaaa gccttatgaa tgttatattt gtcattgctg gtttaccacaa agtggtagca 1260
tgaagatgca cattttacag aagcacacag aaaatgtggc caaatctcac tgtccccact 1320
gtgacacagt catagcccgaa aaaagtgatt tgggtgtcca cttgcgaaag cagcattcct 1380
atattgagca aggcaagaaa tgcggttact gtgatgctgt gtttcatgag cgctatgccc 1440
tcattccagca tcagaagtca cacaagaatg agaagcgtt taagtgtgac cagtgtgatt 1500
acgctttag acaggagagg cacatgatca tgcacaagcg caccacacc ggggagaagc 1560
cttacgcctg cagccactgc gataagacct tccgccagaa gcagcttctc gacatgcact 1620
tcaagcgcta tcacgacccc aacttcgtcc ctgcggcttt tgtctgttct aagtgtggga 1680
aaacatttac acgtcggaat accatggcaa gacatgctga taattgtgct ggcccagatg 1740
gcgtagaggg ggaatatgga ggagaaacga agaagagtaa acgtggaaga aaaagaaaga 1800
tgcgctctaa gaaagaagat tcctctgaca gtgaaaatgc tgaaccagat ctggacgaca 1860
atgaggatga ggaggagcct gccgtagaaa ttgaacctga gccagagcct cagcctgtga 1920
ccccagcccc accaccgcc aagaagcgga gaggacgacc ccctggcaga accaaccagc 1980
ccaaacagaa ccagccaaca gctatcattc aggttgaaga ccagaatata ggtgcaattg 2040
agaacattat agttgaagta aaaaaagagc cagatgctga gcccgagag ggagaggaag 2100
aggaggccca gccagctgcc acagatgccc ccaacggaga cctcacgccc gagatgatcc 2160
tcagcatgat ggaccgg 2177

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<210> SEQ ID NO 61
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 61

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Tyr Tyr Tyr Asn Cys Ala Gly Gly
1          5

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<210> SEQ ID NO 62
<211> LENGTH: 8
<212> TYPE: RNA
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 62

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ggguragu

```

8

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<210> SEQ ID NO 63
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 63

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ctgccccaaag tcgcagccat ggctacaggc tcccggaag

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39

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<210> SEQ ID NO 64
 <211> LENGTH: 41
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 64

gcttatcatg tctggccagc tagctagaag ccacagctgc c

41

What is claimed is:

1. An expression vector for amplified expression in mammalian cells, comprising:

one or more inserted foreign DNA sequences, including human growth hormone (hGH) and a plurality of therapeutic genes, wherein the one or more inserted foreign DNA sequences are fused with intracellular or extracellular secretion sequences;

a promoter, configured to drive expression of the one or more inserted foreign DNA sequences;

an insulator, configured for stable and safety expression of the one or more inserted foreign DNA sequences;

a marker gene, configured to encoding one or more corresponding proteins to indicate the expression of the one or more inserted foreign DNA sequences; and

a regulator element, configured to control expression of the hGH and the plurality of therapeutic genes under a controllable condition.

2. The expression vector according to claim 1, wherein the expression vector includes:

an adenoviral associate virus (AAV) vector and/or a lentiviral vector, the lentiviral vector including a human immunodeficiency virus (HIV) vector; and/or
 a non-viral vector, including a lipid nanoparticle (LNP) vector.

3. The expression vector according to claim 1, wherein: the promoter includes a cytomegalovirus (CMV) promoter or an elongation factor (EF-1) promoter or a synthetic specific promoter.

4. The expression vector according to claim 1, wherein: the marker gene includes a green fluorescent protein (GFP) gene and/or a non-toxic mammalian biomarker.

5. The expression vector according to claim 1, further including:

a neuronal promoter, including human synapsin, configured to drive expression of transgenes in neurons in a central nervous system (CNS).

6. The expression vector according to claim 1, wherein: the insulator includes one or more of a first chromatin insulator, a second chromatin insulator, and a third chromatin insulator, wherein:

the first chromatin insulator includes at least a part of a sequence of:

GAGCTCACGGGACAGCCCCCCCCCAAGCCCCAGGGA
 TGTAATTACGTCCTCCCCCGCTAGGGGCGAGCAGCGAG
 CCGCCCGGGGCTCCGCTCCGGTCCGGCGCTCCCCCGCA
 TCCCCGAGCCGGCAGCGTGCGGGACAGCCGGGCACGG

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GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTC
 GCTGCTCTTTGAGCCTGCAGACACCTGGGGGGATACGGG
 GAAAAAGCTTTAGGCTGAAAGAGAGATTTAGAATGACAG
 GCGCGCCTGGCCATACATCGATACGGTACCGAGTTGGCG
 CGCCTGGGAGCTCACGGGGACAGCCCCCCCCCAAGCCC
 CCAGGGATGTAATTACGTCCTCCCCCGCTAGGGGGCAG
 CAGCGAGCCGCCGGGCTCCGCTCCGGTCCGGCGCTCC
 CCCCCGATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCG
 GGCACGGGGAAGGTGGCACGGGATCGCTTTCCTCTGAAC
 GCTTCTCGTGCTCTTTGAGCCTGCAGACACCTGGGGGG
 ATACGGGGAAAAAGCTT;

the second chromatin insulator includes at least a part of a sequence of:

CACAGGTGCTAATATTCAATTATAGCTCTAAATCAGTGG
 CCAAAAGCGTGGCCCCGACAAAAGGCTTCTATGGGCTTGA
 TTGGAGTAGGCACAGCATCTGAAGTCTATCAAAGTGCTAC
 AATTTTACATCGTTTAAAGACAGATGGGCGAGGAAGTCACC
 ATCCAACCTATGATTACTTTCATTCCCATTAATTTGTAGG
 GGAGAGATTTGTTACAGCAATGGGGTACAGAAATCACCAT
 TCCCTCTGCAATTTACAGTCTGAGAGCCAAAGATGATG
 ACCAAATGGGATATATTCTGGGAAAGGATTAGGAAAAA
 AATGAGGACGGAATAACCCAGCCGATTGAGGTCTCAGTTA
 AATTTGACCGAAGAGGAATTGGATATCCTTCTTAGGTGCG
 GCCACTGTTGAGCCTCCAAGGCCATTGCATTAACACGGA
 AAACCTCAAAACCTGTTTGGGTTGATCAGTGGCCGCTTCC
 TCAACAAAACTGGAGGCATTACATTATGCGAAAGAA
 CAATTAGACAAAGGACATATTGAACCATTTTTTACCATT
 GGAACGCCCCAGTTTTTGTAAATTCAGAAAAAATTCGCTAG
 ATGGCGCATATTGACCGACTTATGAGCGGTTAACGCGACC
 ATTCAACCGATGGGAGCTCTCAACACGAGGTGGCCATCTC
 CGGCCATGATCCCAAAGAAATGGCCGATGGTAATAATTGA
 TCTAAAGATTGTGTTTTTACTATTCTTTGGCTCCTCA

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GGATTTTGAGAAATTTGCATTACAAATACCTGCCATGAAT
 AACAAAGGAACCAGCTACTATATATCGATGGAAAGTTGTAC
 CCCAAGGAATGTTAAATAGTCAAACCTTTTGTGGAAAGGT
 TATTCAGCTTGTGAGGATCAGTTTCCTGATTGCTATATT
 ATTCAATTATGTTGATGACATTTTATGTGCTGCTGCAAGCA

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GGGACAGACGGATTGAGTGTTTTGCTAGGCTACAAGAAGT
 GGTGGGACTTACAGGTCTCGCTATAGCACCAGATAAAATC
 CATACTACACCTATCATTACCTGGGAATGAG;

and

the third chromatin insulator includes at least a part of a
 sequence of:

GAAGGTGA TGCAGTCGAA GCCATTGTGG AGGAGTCCGA
 AACTTTTATTAAAGGAAAGG AGAGAAAGAC TTACCAGAGA CGCCGGGAAG
 GGGGCCAGGA AGAAGATGCC TGCCACTTAC CCCAGAACCAGACGGATGGG
 GGTGAGGTGG TCCAGGATGT CAACAGCAGT GTACAGATGG TGATGATGGA
 ACAGCTGGAC CCCACCTTC TTCAGATGAA GACTGAAGTA ATGGAGGGCA
 CAGTGGCTCC AGAAGCAGAG GCTGCTGTGG ACGATACCCA GATTATAACT
 TTACAGGTTG TAAATATGGA GGAACAGCCC ATAAACATAG GAGAACTTCA
 GCTTGTTCAA GTACCTGTTT CTGTGACTGT ACCTGTTGCT ACCACTTCAG
 TAGAAGAACT TCAGGGGGCT TATGAAATG AAGTGTCTAA AGAGGGCCTT
 GCGGAAAGTG AACCCATGAT ATGCCACACC CTACCTTTGC CTGAAGGGTT
 TCAGGTGGTT AAAGTGGGGG CCAATGGAGA GGTGGAGACA CTAGAACAAG
 GGGAACTTCC ACCCCAGGAA GATCCTAGTT GGCAAAAAGA CCCAGACTAT
 CAGCCACCAG CCAAAAAAAC AAAGAAAACCAAAAAGAGCA AACTGCGTTA
 TACAGAGGAG GGCAAAGATG TAGATGTGTC TGTCTACGAT TTTGAGGAAG
 AACAGCAGGA GGGTCTGCT TCAGAGGTGA ATGCAGAGAA AGTGGTTGGT
 AATATGAAGC CTCCAAGCC AACAAAAATT AAAAAGAAAG GTGTAAGAA
 GACATTCCAG TGTGAGCTTT GCAGTTACAC GTGTCCACGG CGTTCAAATT
 TGGATCGTCA CATGAAAAGC CACACTGATG AGAGACCACA CAAGTGCCAT
 CTCTGTGGCA GGGCATTGAG AACAGTCACC CTCCTGAGGA ATCACCTTAA
 CACACACACA GGTACTCGTC CTCACAAGTG CCCAGACTGC GACATGGCCT
 TTGTGACCAG TGGAGAATTG GTTCGGCATC GTCGTTACAA ACACACCCAC
 GAGAAGCCAT TCAAGTGTTT CATGTGCGAT TACGCCAGTG TAGAAGTCAG
 CAAATTAAAA CGTCACATTC GCTCTCATAC TGGAGAGCGT CCGTTTCAGT
 GCAGTTTGTG CAGTTATGCC AGCAGGGACA CATAACAAGCT
 GAAAAGGCACATGAGAACC ATTCAAGGGA AAAGCCTTAT GAATGTTATA
 TTTGTCATGC TCGGTTTACC CAAAGTGGA CCATGAAGAT GCACATTTTA
 CAGAAGCACA CAGAAAATGT GGCCAAATTT CACTGTCCCC ACTGTGACAC
 AGTCATAGCC CGAAAAAGTG ATTTGGGTGT CCACTTGCGA AAGCAGCATT
 CCTATATTGA GCAAGGCAAG AAATGCCGTT ACTGTGATGC TGTGTTTCAT
 GAGCGCTATG CCCTCATCCA GCATCAGAAG TCACACAAGA ATGAGAAGCG
 CTTTAAGTGT GACCAGTGTG ATTACGCTTG TAGACAGGAG AGGCACATGA
 TCATGCACAA GCGCACCCAC ACCGGGGAGA AGCCTTACGC CTGCAGCCAC

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TGCGATAAGA CCTTCCGCCA GAAGCAGCTT CTCGACATGC ACTTCAAGCG
 CTATCACGAC CCCAACTTCG TCCCTGCGGC TTTTGTCTGT TCTAAGTGTG
 GGAAACATT TACACGTCGG AATACCATGG CAAGACATGC TGATAATTGT
 GCTGGCCAG ATGGCGTAGA GGGGGAAAAT GGAGGAGAAA CGAAGAAGAG
 TAAACGTGGA AGAAAAAGAA AGATGCGCTC TAAGAAAGAA GATTCTCTG
 ACAGTGAAAA TGCTGAACCAGATCTGGACG ACAATGAGGA TGAGGAGGAG
 CCTGCCGTAG AAATTGAACC TGAGCCAGAGCCTCAGCCTG TGACCCAGC
 CCCACCACCC GCCAAGAAGC GGAGAGGACG ACCCCCTGGCAGAACCAACC
 AGCCCAACA GAACCAGCCA ACAGCTATCA TTCAGGTTGA AGACCAGAA
 ACAGGTGCAA TTGAGAACAT TATAGTTGAA GTAAAAAAG AGCCAGATGC
 TGAGCCCGCA GAGGGAGAGG AAGAGGAGGC CCAGCCAGCT GCCACAGATG
 CCCCCAACGG AGACCTCAG CCCGAGATGA TCCTCAGCAT GATGGACCGG

7. The expression vector according to claim 2, wherein: a surface capsule of the expression vector is engineered by inserting a nucleotide sequence to bind and/or recognize one or more receptors on Choroid plexus and blood brain barrier (BBB) tissues.

8. The expression vector according to claim 2, wherein: a surface capsule of the expression vector is engineered by inserting a single-chain fragment variable (scFv) antibody sequence to bind and/or recognize one or more receptors on Choroid plexus and blood brain barrier (BBB) tissues.

9. The expression vector according to claim 2, wherein: a surface capsule of the expression vector is engineered by inserting a ligand sequence derived from the hGH to bind and/or recognize one or more hGH receptors on Choroid plexus and blood brain barrier (BBB) tissues.

10. The expression vector according to claim 1, wherein: a biological tracer is used as an investigative and/or diagnostic tool to mark cells and/or to generate a specific animal model in vivo, and/or the biological tracer is applied for diagnostic instruments.

11. The expression vector according to claim 1, wherein: the one or more inserted foreign DNA sequences are genes encoding proteins, peptides and transcription factors including IGF-1 (insulin growth factor-1), insulin, rat growth hormone, glucocerebrosidase, brain-derived neurotrophic factor (BDNF), β -endorphin, HSV-1 (herpes simplex virus), anti-glioma, ApoE4, presenilin 1 and/or 2, myelin oligodendrocyte glycoprotein (MOG), crry, gata4, glutamate decarboxylase 65, interleukin-10, chondroitinase ABC, P/NK-1R, human interleukin-12 (hIL-12), stem cell transcription factors for iPSC including Oct4, Sox2, Klf1-5, Nanog, LIN28 and Glis1, direct dopaminergic neurons reprogramming including ASC11, LMX1A, MYT1L and BRN2, direct glutamatergic, GABAergic neuronal reprogramming including ASC11, MYT1L, BRN2, and NeuroD1, direct retinal pigment epithelium reprogramming including PAX6, RAX, CRX, MITF-A, OTX2, NRL and KLF4/2, and direct spinal motor neurons reprogramming including NGN2, SOX11, ISL1 and LHX3, and further including anti-Tau, anti- β -amyloid antibodies, and nanobodies (VHH).

12. The expression vector according to claim 1, wherein: the one or more inserted foreign DNA sequences include CRISPR/Cas9, antisense, sense, or mRNA used for treating a central nervous system (CNS) disease.

13. The expression vector according to claim 1, wherein: the one or more inserted foreign DNA sequences include antisense or sense used for treating a central nervous system (CNS) disease, the CNS disease including brain viral infection.

14. The expression vector according to claim 1, wherein: the one or more inserted foreign DNA sequences include shRNA, RNAi, or mRNA used for treating a central nervous system (CNS) disease, the CNS disease including brain viral infection.

15. The expression vector according to claim 1, wherein: the expression vector is used to treat a central nervous system (CNS) disease including brain viral infection by delivering the hGH and the plurality of therapeutic genes via crossing the BBB.

16. The expression vector according to claim 15, wherein: delivering the hGH and the plurality of therapeutic genes into the CNS via crossing the BBB is administrated with intravenous injection formulation, nasal spray formulation, inhalation formulation, liquid and/or tablet formulations, eye drop administration formulation, or a combination thereof.

17. The expression vector according to claim 1, wherein the regulator element includes:

(1) a tetracycline response element (TRE) including sequences of "TCCCTATCAGTGATAGAGA" to be included into the vector; and/or

(2) a post-transcriptional regulatory element (PRE)+derived from woodchuck hepatitis virus (WHV) into 3' untranslated region of the AAV vector or lentiviral vector to enhance both titer and transgene expression.

18. The expression vector according to claim 1, wherein: the intracellular or extracellular secretion sequences include fusion sequences, the fusion sequences including ROIKIWFORRR, YGRKKRRORRR, and/or LLNFDLLKILLAGDWESNPCP.

19. A method for generating an expression vector for amplified expression in mammalian cells, comprising:

fusing one or more inserted foreign DNA sequences with intracellular or extracellular secretion sequences, the one or more inserted foreign DNA sequences including human growth hormone (hGH) and a plurality of therapeutic genes;

configuring a promoter to drive expression of the one or more inserted foreign DNA sequences;

configuring an insulator for stable and safety expression of the one or more inserted foreign DNA sequences;

configuring a marker gene to encoding one or more corresponding proteins to indicate the expression of the one or more inserted foreign DNA sequences; and

configuring a regulator element control expression of the hGH and the plurality of therapeutic genes under a controllable condition.

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